

Information Security and Cyber Laws

Distributed Information Systems, Role of Internet and Web services, Threats and attacks, **Assessing-Damages**, Security in Mobile and Wireless Computing, Security Threats to E-Commerce, E—Governance and EDI, Concepts in Electronics payment systems, E-Cash, Credit/Debit Cards. Physical -Needs Disaster and Controls, Basic Tenets of Physical Security and Physical Entry Controls, Access Control. Model of Cryptographic Systems, Design and Implementation Issues, Policies, **Network Security, Attacks**, Need of Intrusion 'Monitoring and Detection, Intrusion Detection. **Security metrics- Classification and their benefits**. Information Security & Laws, Ethics-Ethical Issues, Issues in Data and Software Privacy. Overview and types of Cyber Crimes.

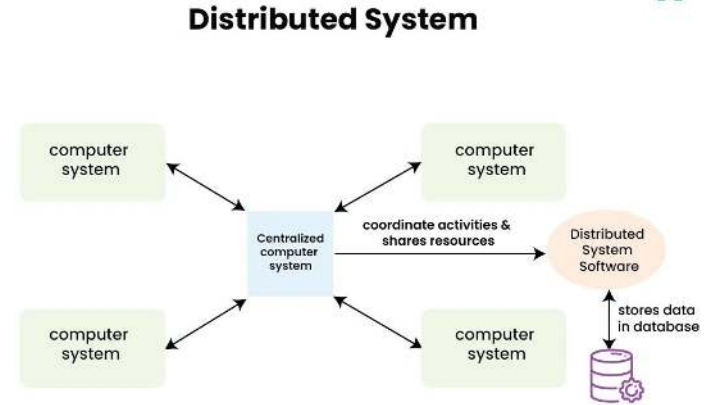
Chapter - 1

Distributed Information Systems



Distributed Information Systems in Distributed System

Distributed systems consist of multiple independent computers working together as a unified system. These systems offer enhanced **scalability, reliability, and fault tolerance**. They ensure efficient resource utilization and performance by distributing tasks across various nodes. One crucial aspect is managing and sharing data effectively, which is the role of distributed information systems.



Fundamentals of Distributed Information Systems

Distributed information systems consist of multiple independent computers working

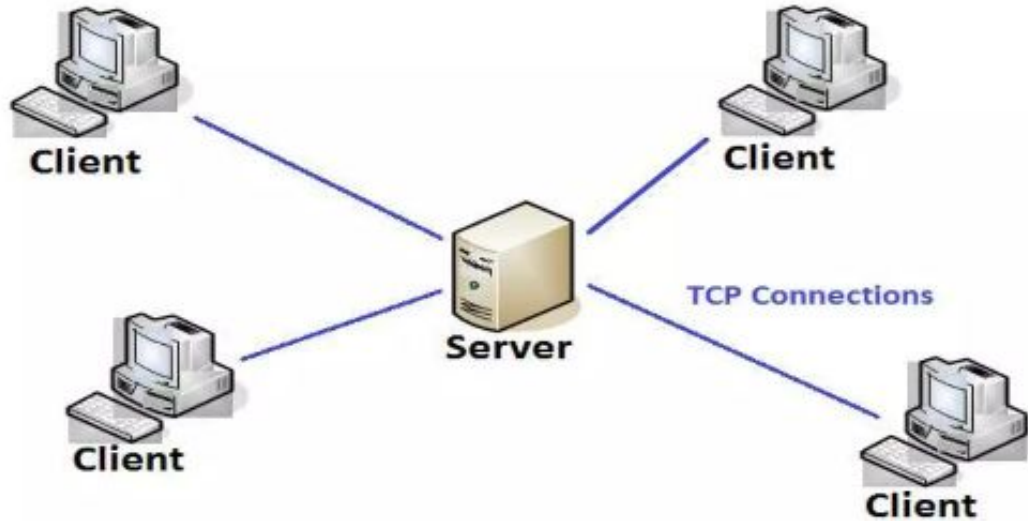
Key Characteristics of Distributed Information Systems:

Scalability: Distributed systems can easily scale by adding more nodes to handle increased load. This allows the system to accommodate growth without performance degradation.

- **Fault Tolerance:** The system continues to operate despite failures of individual nodes. Redundancy and data replication ensure that a failure does not impact the overall functionality.
- **Concurrency:** Multiple nodes operate simultaneously, performing tasks in parallel. This improves the system's overall efficiency and speed.
- **No Global Clock:** There is no single time reference across nodes, which can lead to synchronization challenges. Distributed systems use logical clocks to manage time.
- **Independent Failures:** Nodes can fail independently without affecting the entire system. This isolation of failures helps maintain system stability.
- **Geographical Distribution:** Nodes can be distributed across different locations. This allows for data localization and reduces latency for users in various regions.
- **Data Consistency:** Ensuring consistency across all nodes is challenging but essential. Distributed systems use protocols to maintain data integrity and consistency.

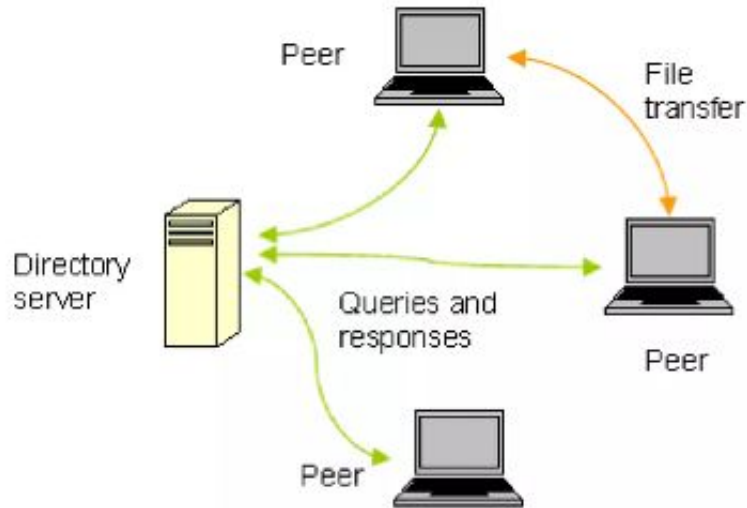
Common Architectural Models are:

- **Client-Server Model:** In this model, clients request services, and servers provide them. This setup is straightforward and easy to implement, making it widely used.



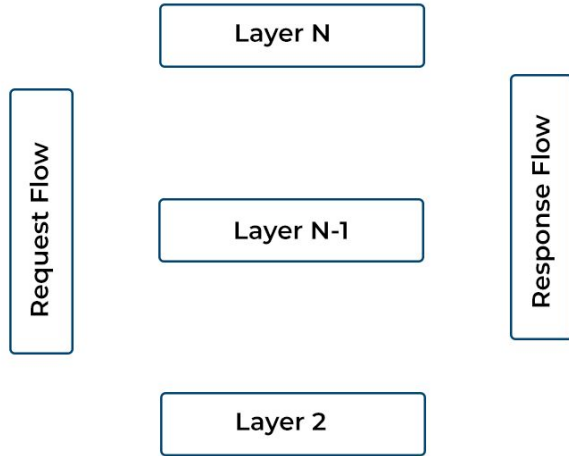
Common Architectural Models are:

Peer-to-Peer Model: All nodes act as both clients and servers. This model is decentralized and offers better fault tolerance and load distribution.



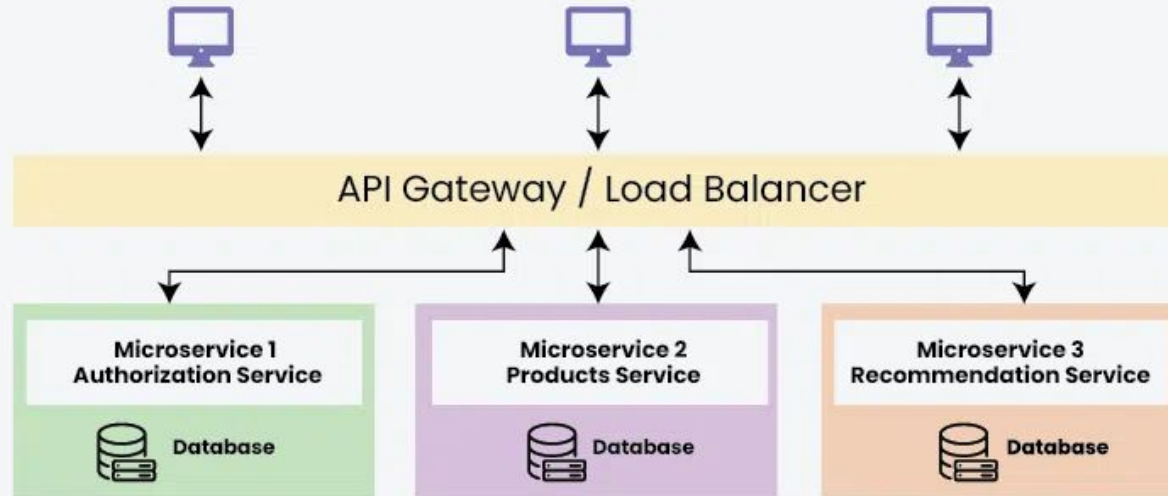
- **Layered Model:** The system is divided into layers, each with specific responsibilities. This separation of concerns improves maintainability and scalability.

LAYERED ARCHITECTURE STYLE



- **Microservices Architecture:** The application is divided into small, independent services. Each service can be developed, deployed, and scaled independently.

Microservices Architecture for Distributed Systems



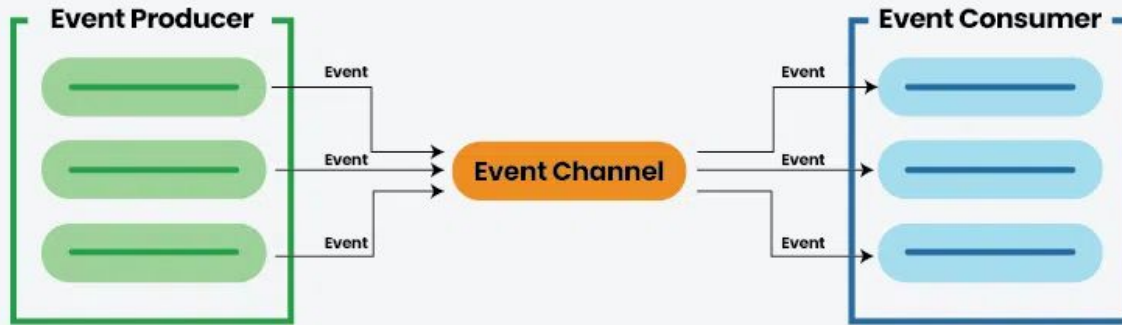
Service-Oriented Architecture (SOA): Services communicate over a network, often using web services. This model promotes reusability and integration of heterogeneous systems.

Service-Oriented Architecture (SOA) in Distributed Systems



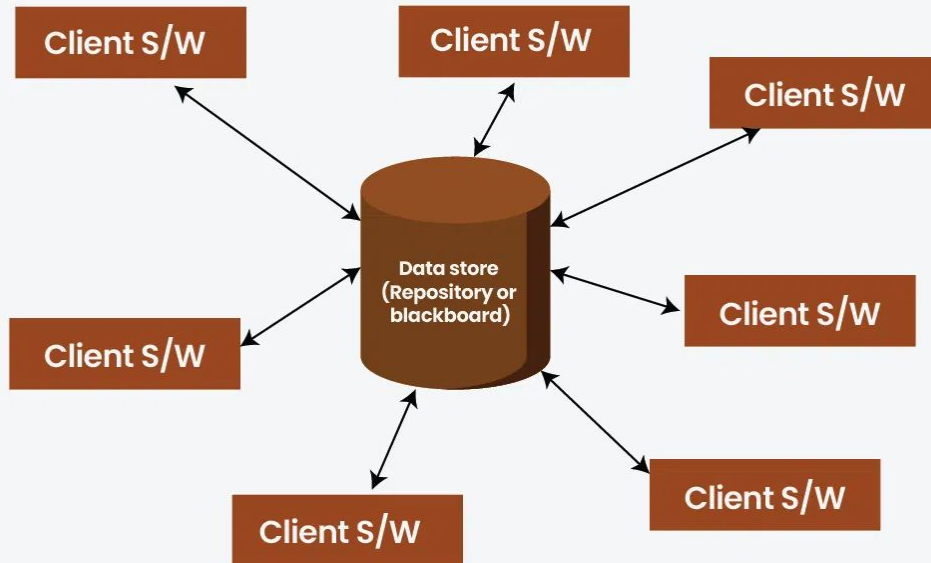
Event-Driven Architecture: Components communicate through events, enabling asynchronous processing. This model is highly responsive and scalable, suitable for real-time applications.

Event-Based Architecture in Distributed Systems



Repository Model: All components interact with a central data repository. This model simplifies data management and ensures data consistency.

Data-Centric Architecture in Distributed Systems



A distributed system is :

- (a) A collection of possibly heterogeneous computer systems that are networked to provide the users with access to the various resources that the system maintains.
- b) A collection of physically separate computer systems that are networked to provide the users with access to the various resources that the system maintains.
- (c) A collection of physically separate, possibly heterogeneous computer systems that are networked to provide the users with access to the various resources that the system maintains.
- (d) A collection of computer systems that are networked to provide the users with access to the various resources that the system maintains.

In a Distributed Information System, replication is used mainly to improve:

- A) Power consumption
- B) Data redundancy and availability
- C) Keyboard input speed
- D) Hardware design standards

Answer: B

Explanation:

Replication stores copies of data across multiple sites to improve **availability, reliability, and fault tolerance**. It helps the system continue functioning even if one node fails.

A distributed service that allows distributed processes to coordinate with each other through a shared hierarchical name space of data registers is known as:

- (a) Oozie
- (b) Mahout
- (c) ZooKeeper
- (d) Sqoop

Communication Protocols in Distributed Information Systems

Communication protocols are essential for ensuring reliable and efficient data exchange in distributed information systems. These protocols define the rules and conventions for communication between nodes, enabling them to work together seamlessly. Without effective communication protocols, nodes in a distributed system would struggle to coordinate and share information, leading to inefficiencies and potential failures.

Key Communication Protocols are:

- **HTTP/HTTPS:** The HyperText Transfer Protocol (Secure) is widely used for communication over the internet. It ensures secure and reliable data transfer between clients and servers.

- **TCP/IP:** The Transmission Control Protocol/Internet Protocol is fundamental to network communication. It provides reliable, ordered, and error-checked delivery of data.
- **UDP:** The User Datagram Protocol is a simpler alternative to TCP. It offers faster communication but without guarantees of delivery, order, or error checking.
- **gRPC:** This modern open-source protocol enables remote procedure calls between distributed systems. It supports multiple programming languages and efficient data serialization.
- **REST:** Representational State Transfer is a protocol for building web services. It uses standard HTTP methods and is stateless, making it scalable and easy to use.
- **WebSockets:** This protocol provides full-duplex communication channels over a single TCP connection. It is ideal for real-time applications like chat systems and online gaming.
- **AMQP:** The Advanced Message Queuing Protocol is used for message-oriented middleware. It ensures reliable message delivery and supports complex routing.

Role of Internet and Web services

Role of Internet and Web services

Internet and its Services

The Internet provides many services for users to share and access information. We can upload or download files like software, videos, audio, or documents.

- These services make the Internet a major source of knowledge and communication.
- To connect and exchange data, we use Internet services such as FTP, Telnet, and VoIP.

Internet services

To access/exchange a large amount of data such as software, audio clips, video clips, text files, other documents, etc., we need internet services. You must use an Internet service to connect to the Internet. Data can be sent from Internet servers to your machine via Internet service. Some of the commonly used internet services are :

- Communication Services
- Information Retrieval Services
- File Transfer
- World Wide Web Services
- Web Services
- Directory Services
- Automatic Network Address Configuration
- Network Management Services
- Time Services
- Usenet
- NewsGroup
- Ecommerce

Now let us discuss them one by one

1. Communication Services: To exchange data/information among individuals or organizations, we need communication services. Following are some of the common communication services:

- **IRC(Internet Relay Chat):** Subscribers can communicate in real-time by connecting numerous computers in public spaces called channels.
- **VoIP:** It stands for Voice over Internet Protocol, which describes how to make and receive phone calls over the internet. A larger number of people believe VoIP is a viable alternative to traditional landlines. VoIP (Voice over Internet Protocol) is a technique that helps us make voice calls via the Internet rather than over a traditional (or analog) phone line.
- Some VoIP services may let you call only other VoIP users, while others may let you call anyone with a phone number, including long-distance, mobile, and local/international lines.

If you have an internet connection you can easily call anyone without using a local phone service because VoIP solutions are based on open standards, they can be used on any computer. More than just setting up calls is what VoIP service providers do. Outgoing and incoming calls are routed through existing telephone networks by them.

- **List Server (LISTSERV):** Delivers a group of email recipients' content-specific emails.
- **E-Mail:** Used to send electronic mail via the internet. It is a paperless method for sending text, images, documents, videos, etc from one person to another via the internet.
- **User Network (USENET):** It hosts newsgroups and message boards on certain topics, and it is mostly run by volunteers.
- **Telnet:** It's used to connect to a remote computer that's connected to the internet.

- **Video Conferencing:** Video conferencing systems allow two or more people who are generally in different locations to connect live and visually. Live video conferencing services are necessary for simulating face-to-face talks over the internet. The system can vary from very simple to complex, depending on the live video conferencing vendors. A live video-based conference involves two or more individuals in separate locations utilizing video-enabled devices and streaming voice, video, text, and presentations in real-time via the internet. It allows numerous people to connect and collaborate face to face over large distances. Tools available for this purpose are Zoom, FreeConference, Google Hangouts, Skype, etc.

2. Information Retrieval Services: It is the procedure for gaining access to information/data stored on the Internet. Net surfing or browsing is the process of discovering and obtaining information from the Internet. When your computer is linked to the Internet, you may begin retrieving data. To get data, we need a piece of software called a Web browser. A print or computer-based information retrieval system searches for and locates data in a file, database, or other collection of data. Some sites are:

3. File Transfer: The exchange of data files across computer systems is referred to as file transfer. Using the network or internet connection to transfer or shift a file from one computer to another is known as file transfer. To share, transfer, or send a file or logical data item across several users and/or machines, both locally and remotely, we use file transfer. Data files include - documents, multimedia, pictures, text, and PDFs and they can be shared by uploading or downloading them. To retrieve information from the internet, there are various services available such as:

- **Gopher:** A file retrieval application based on hierarchical, distributed menus that is simple to use.
- **FTP (File Transfer Protocol):** To share, transfer, or send a file or logical data item across several users and/or machines, both locally and remotely.
- **Archie:** A file and directory information retrieval system that may be linked to FTP

4. Web services: Web services are software that uses defined messaging protocols and are made accessible for usage by a client or other web-based programs through an application service provider's web server. Web services allow information to be exchanged across web-based applications. Using Utility Computing, web services can be provided.

5. World Wide Web: The internet is a vast network of interconnected computers. Using this network, you can connect to the world wide web (abbreviated as 'www' or 'web') is a collection of web pages. The web browser lets you access the web via the internet.

6. Directory Services: A directory service is a set of software that keeps track of information about your company, customers, or both. Network resource names are mapped to network addresses by directory services. A directory service provides users and administrators with full transparent access to printers, servers, and other network devices. The directory services are :

- **DNS (Domain Number System):** This server provides DNS. The mappings of computer hostnames and other types of domain names to IP addresses are stored on a DNS server.
- **LDAP (Lightweight Directory Access Protocol):** It is a set of open protocols that are used for obtaining network access to stored data centrally. It is a cross-platform authentication protocol for directory services and also allows users to interact with other directory services servers.

7. Automatic Network Address Configuration: Automatic Network Addressing assigns a unique IP address to every system in a network. A DHCP Server is a network server that is used to assign IP addresses, gateways, and other network information to client devices. It uses Dynamic Host Configuration Protocol as a common protocol to reply to broadcast inquiries from clients.

8. Network Management Services: Network management services are another essential internet service that is beneficial to network administrators. Network management services aid in the prevention, analysis, diagnosis, and resolution of connection problems. The two commands related to this are:

- **ping:** The ping command is a Command Prompt command that is used to see if a source can communicate with a specific destination & get all the possible paths between them.
- **tracert:** To find the path between two connections, use the tracert command.

11. News Group: It is a lively Online Discussion Forum that is easily accessible via Usenet. Each newsgroup contains conversations on a certain topic, as indicated by the newsgroup name. Users can use newsreader software to browse and follow the newsgroup as well as comment on the posts. A newsgroup is a debate about a certain topic made up of notes posted to a central Internet site and distributed over Usenet, a global network of news discussion groups. It uses Network News Transfer Protocol (NNTP).

12. E-commerce: Electronic commerce, also known as e-commerce or e-Commerce, is a business concept that allows businesses and individuals to buy and sell goods through the internet. Example: Amazon, Flipkart, etc. websites/apps.

Threats and attacks

Threats to Information Security :

Information security threats represent actions or events that can compromise the confidentiality, integrity, or availability of data and systems. These threats originate from various sources, including individuals, organised groups, or even natural disasters.



Understanding Information Security Principles

Information security programs are built around three core objectives, commonly known as the CIA (Confidentiality, Integrity, Availability):

- **Confidentiality:** Ensures information remains accessible only to authorised individuals, entities, and processes. A breach occurs when unauthorised parties gain access to sensitive data, such as passwords or personal information.
- **Integrity:** Maintains the accuracy and completeness of data by preventing unauthorised modifications. This principle ensures that only authorised personnel can edit or update information, maintaining data reliability.
- **Availability:** Guarantees that information and systems remain accessible when needed. This requires coordination between network operations, development teams, incident response, and policy management to prevent service disruptions.

Common Information Security Threats

There are many different risks that can harm your data and systems. Knowing about these threats helps you protect yourself better.

Malware Threats Malware is malicious software made to damage or take over your computer. It can spread fast and cause big problems.

- Viruses are self-replicating programs that attach themselves to other programs or files on host computers. They spread through infected media like songs, videos, or software downloads. Notable examples include file viruses, macro viruses, boot sector viruses, and stealth viruses.

Malware

Malware (short for “malicious software”) is a file or code, typically delivered over a network, that infects, explores, steals or conducts virtually any behavior an attacker wants. And because malware comes in so many variants, there are numerous methods to infect computer systems.

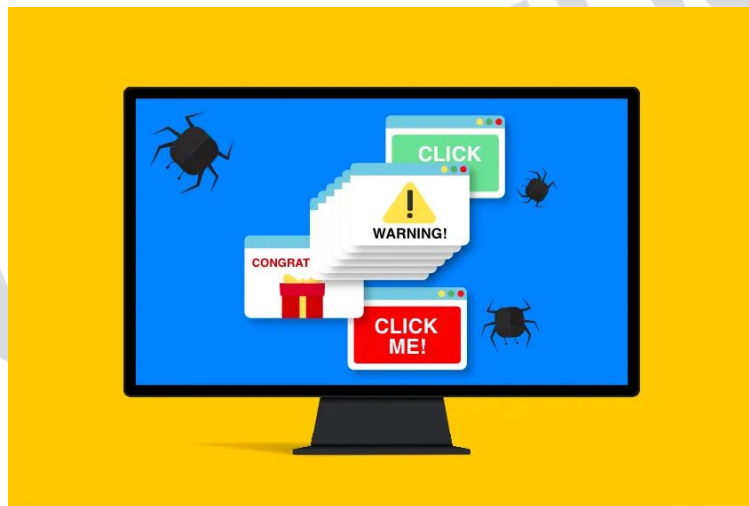
Types of malware



Types of Malware:

Malware is an inclusive term for all types of malicious software. Malware examples, malware attack definitions and methods for spreading malware include:

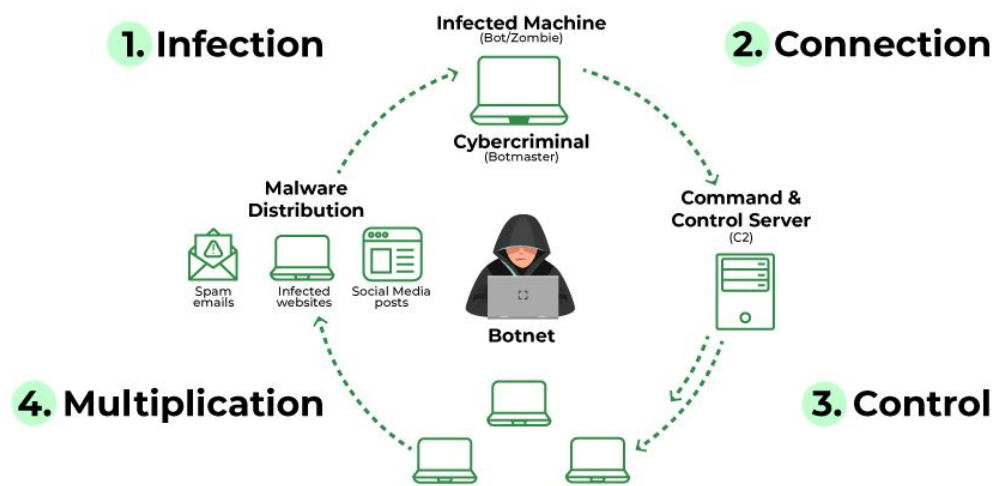
Adware – While some forms of adware may be considered legitimate, others make unauthorized access to computer systems and greatly disrupt users.



Botnets –

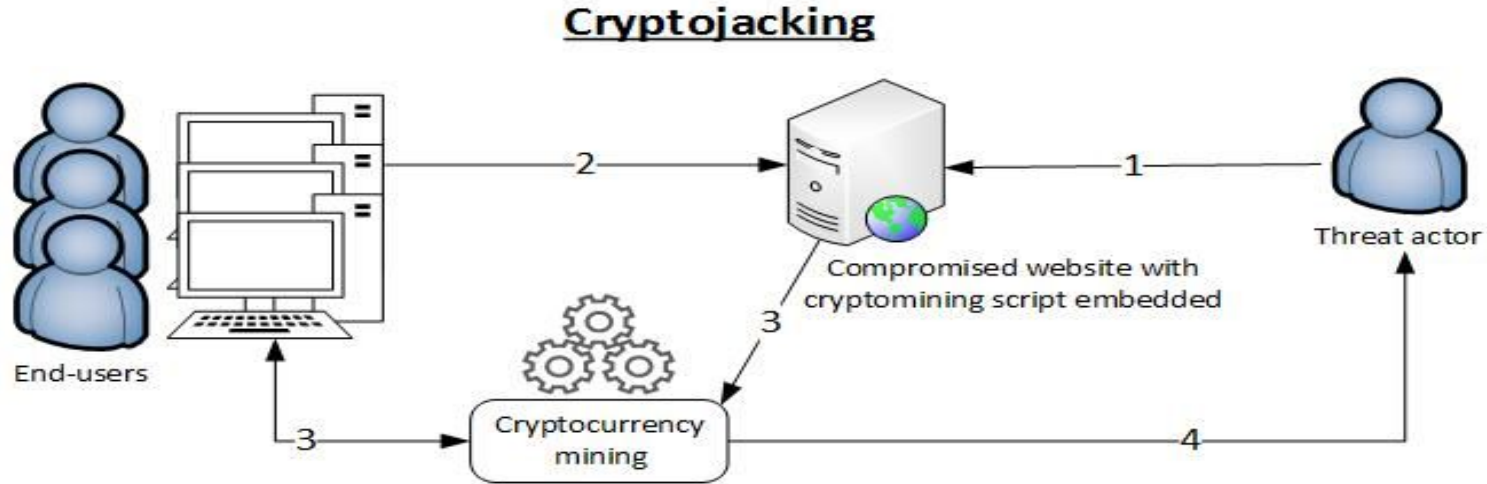
Short for “robot network,” these are networks of infected computers under the control of single attacking parties using command-and-control servers. Botnets are highly versatile and adaptable, able to maintain resilience through redundant servers and by using infected computers to relay traffic. Botnets are often the armies behind today's distributed denial-of-service (DDoS) attacks.

How a Botnet works



Cryptojacking –

is malicious cryptomining (the process of using computing power to verify transactions on a blockchain network and earning cryptocurrency for providing that service) that happens when cybercriminals hack into both business and personal computers, laptops, and mobile devices to install software.

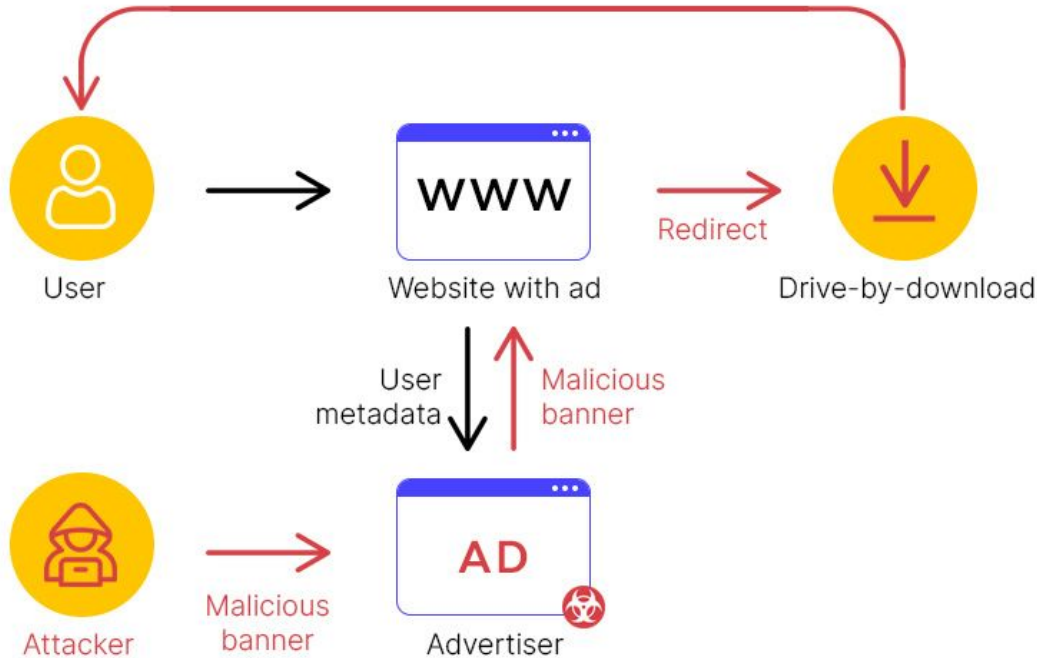


Steps

1. The threat actor compromises a website
2. Users connect to the compromised website and the cryptomining script executes
3. Users unknowingly start mining cryptocurrency on behalf of the threat actor
4. Upon successfully adding a new block to the blockchain, the threat actor receives a reward in cryptocurrency coins

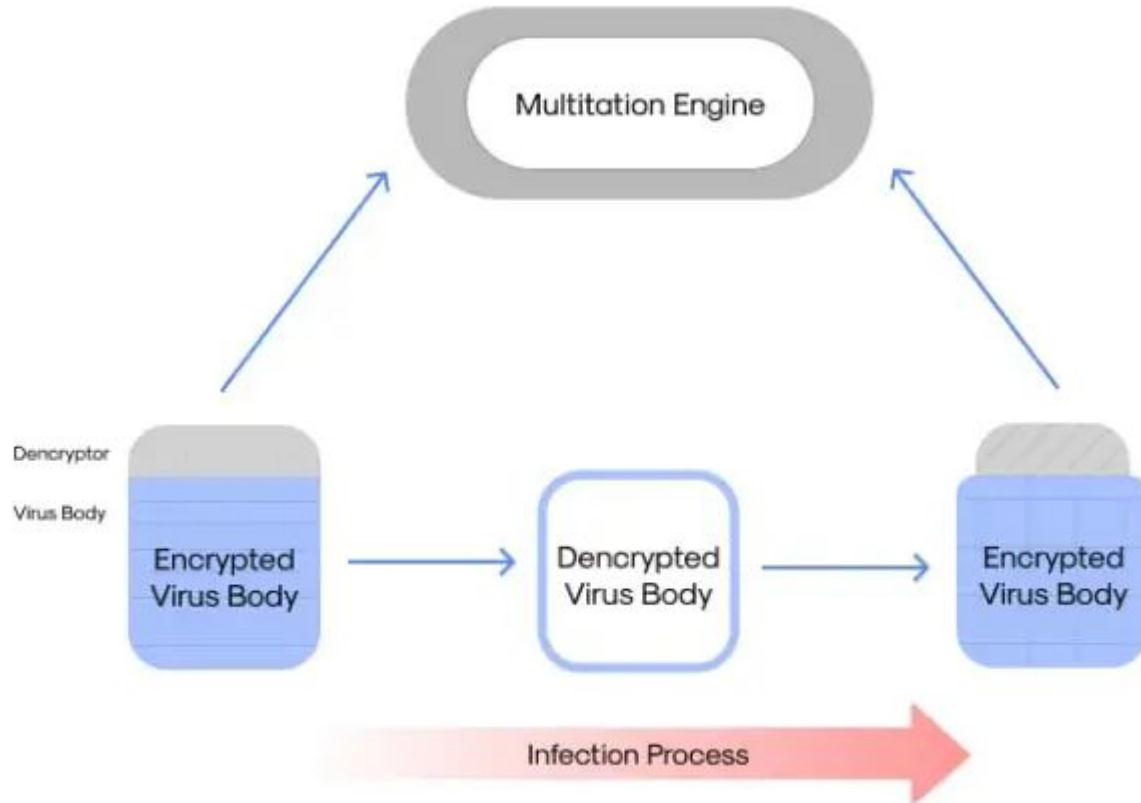
Malvertising –

Malvertising is a portmanteau of “malware + advertising” describing the practice of online advertising to spread malware. It typically involves injecting malicious code or malware-laden advertisements into legitimate online advertising networks and webpages.



Polymorphic malware –

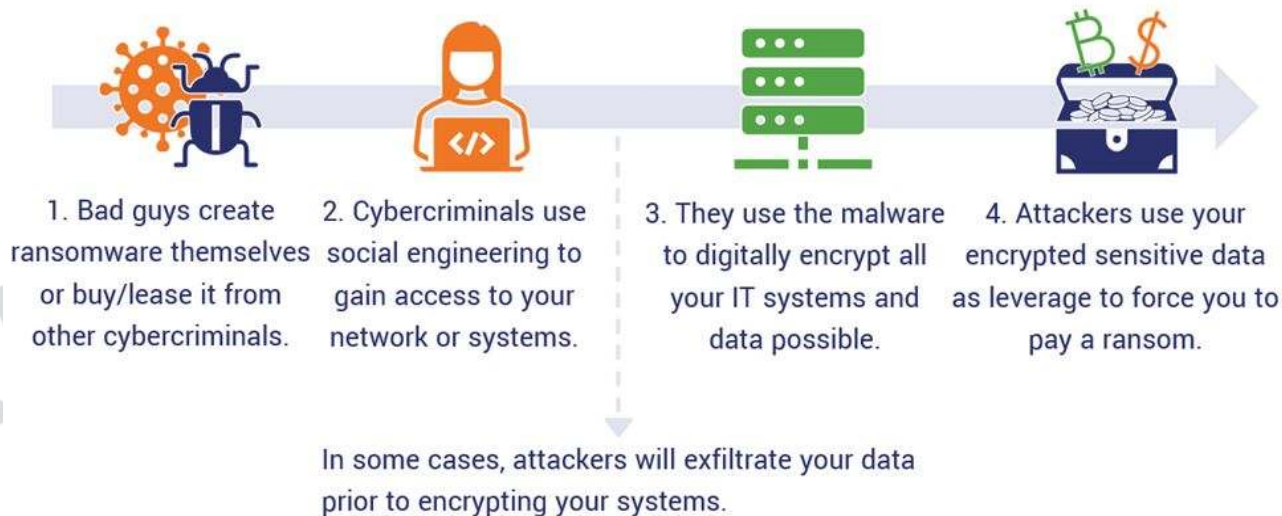
Any of the above types of malware with the capacity to “morph” regularly, altering the appearance of the code while retaining the algorithm within.



Ransomware –

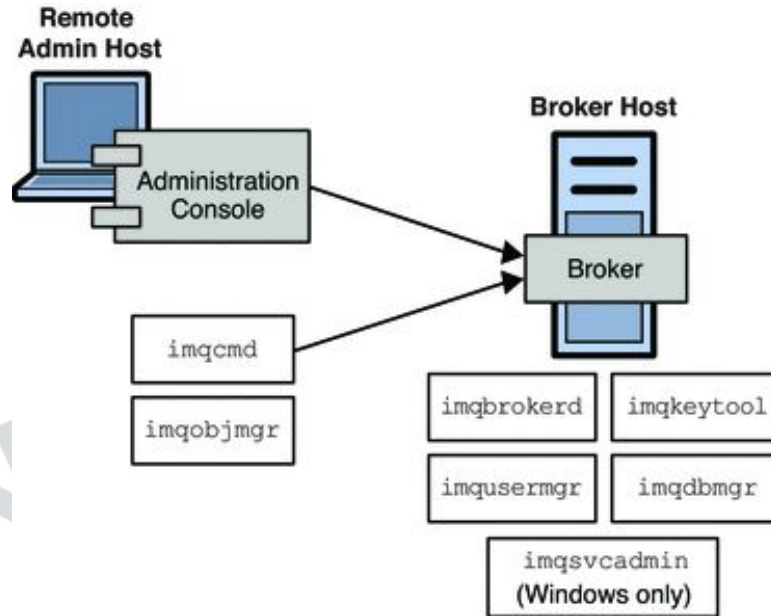
Is a criminal business model that uses malicious software to hold valuable files, data or information for ransom. Victims of a ransomware attack may have their operations severely degraded or shut down entirely.

How Ransomware Works



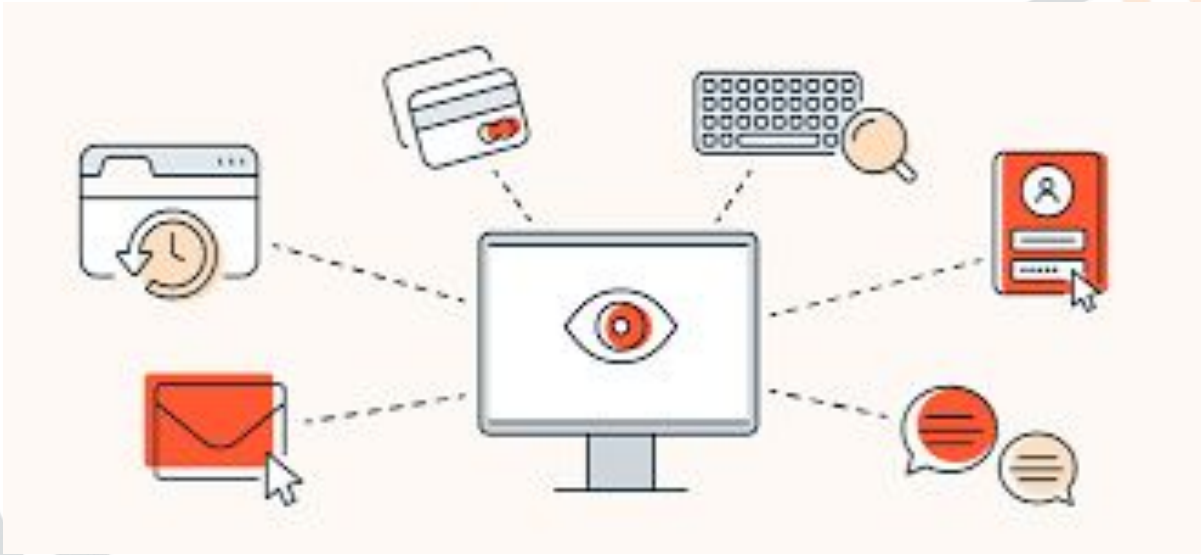
Remote Administration Tools (RATs) –

Software that allows a remote operator to control a system. These tools were originally built for legitimate use, but are now used by threat actors. RATs enable administrative control, allowing an attacker to do almost anything on an infected computer.



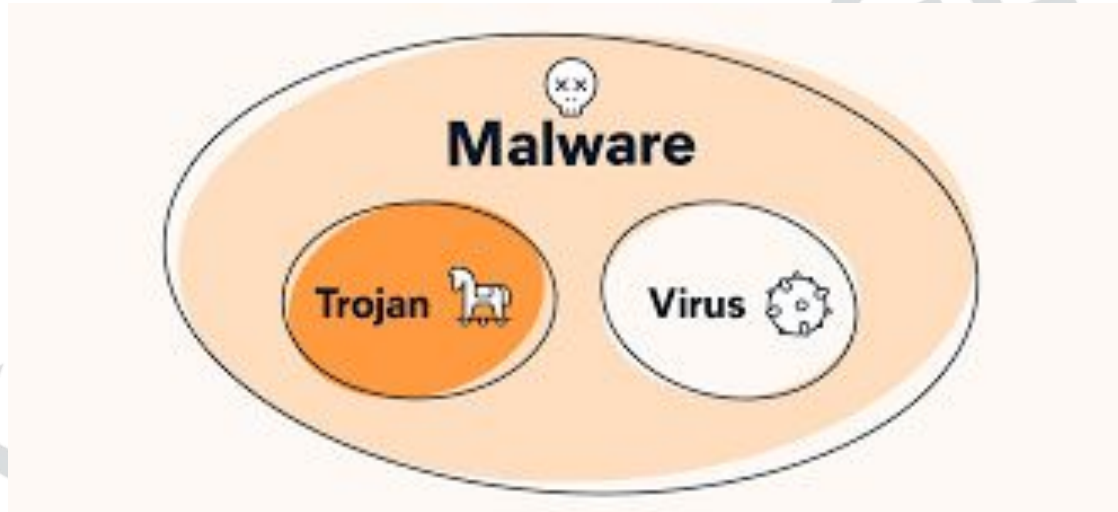
Spyware –

Malware that collects information about the usage of the infected computer and communicates it back to the attacker. The term includes botnets, adware, backdoor behavior, keyloggers, data theft and net-worms.



Trojans Malware –

Malware disguised in what appears to be legitimate software. Once activated, malware Trojans will conduct whatever action they have been programmed to carry out. Unlike viruses and worms, Trojans do not replicate or reproduce through infection. “Trojan” alludes to the mythological story of Greek soldiers hidden inside a wooden horse that was given to the enemy city of Troy.



Virus Malware –

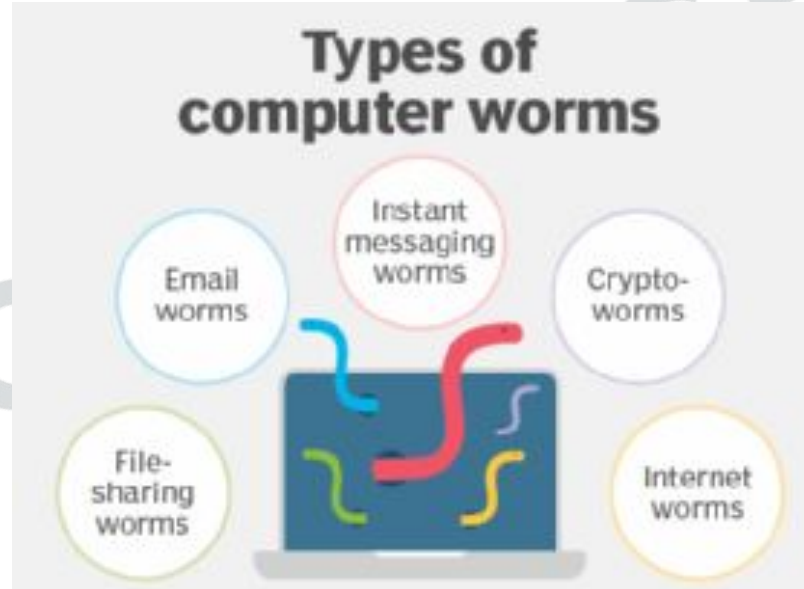
Programs that copy themselves throughout a computer or network. Malware viruses piggyback on existing programs and can only be activated when a user opens the program. At their worst, viruses can corrupt or delete data, use the user's email to spread, or erase everything on a hard disk.

Types of viruses

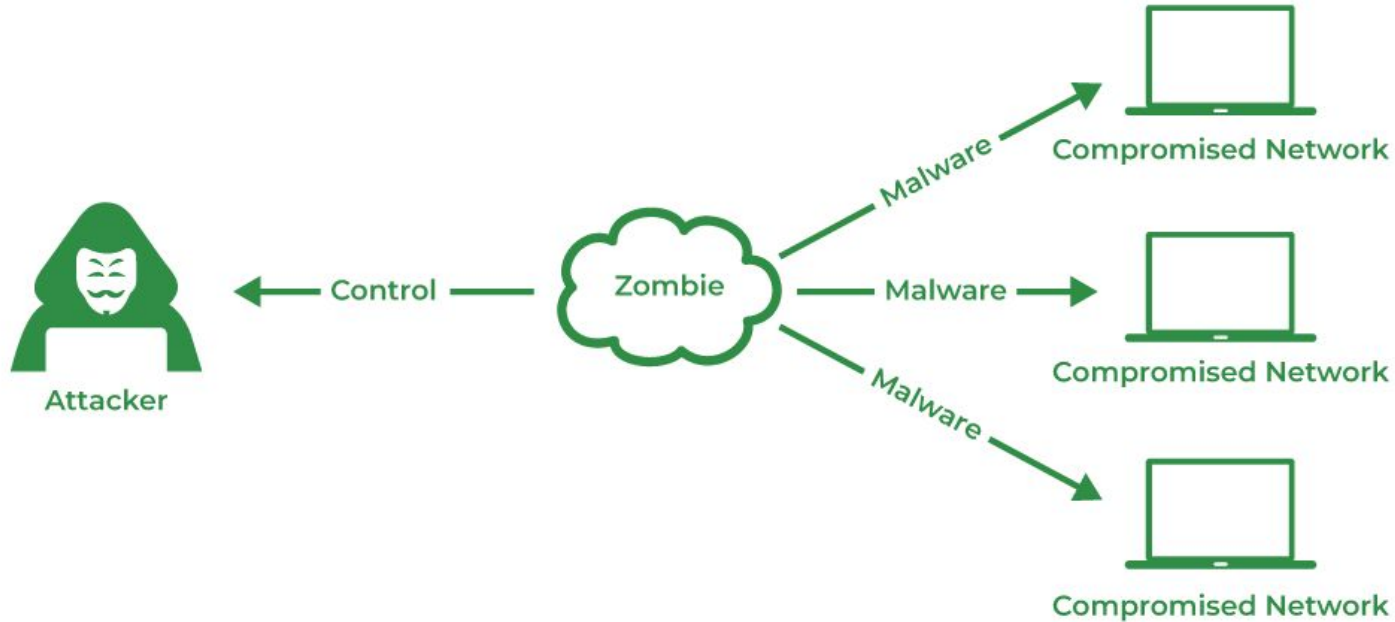


Worm Malware –

Self-replicating viruses that exploit security vulnerabilities to automatically spread themselves across computers and networks. Unlike many viruses, malware worms do not attach to existing programs or alter files. They typically go unnoticed until replication reaches a scale that consumes significant system resources or network bandwidth.



- **Zombies** operate similarly to spyware but focus on awaiting commands from hackers rather than stealing information. These infected systems can be remotely controlled to participate in coordinated attacks.



- **Scareware** masquerades as legitimate system repair tools but actually infects or destroys systems upon execution. These programs display frightening messages to pressure users into paying for fake system fixes.



- **Rootkits** are designed to gain administrative privileges within user systems. Once root access is obtained, attackers can steal private files, access sensitive data, or maintain persistent system control.



A salami attack is a method of cybercrime that attackers or a hacker typically used to commit financial crimes. Cybercriminals steal money or resources from financial accounts on a system one at a time. This attack occurs when several minor attacks combine to create a sturdy attack. because of this sort of cybercrime, these attacks frequently go undetected. Salami attacks are used for the commission of economic crimes Those who are found guilty of such an attack face punishment under Section 66 of the IT Act.



A trapdoor virus (or backdoor) is a hidden method, a secret entry point, in software or a system that bypasses normal security, allowing unauthorized access, often planted by developers or attackers for later exploitation, like a secret password or command.



Which of the following can be used to prevent improper authentication due to password exposure?

- (a) Trapdoor
- (b) One-time Password
- (c) Trojan Horse
- (d) Anti-virus software

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Ans. (a)

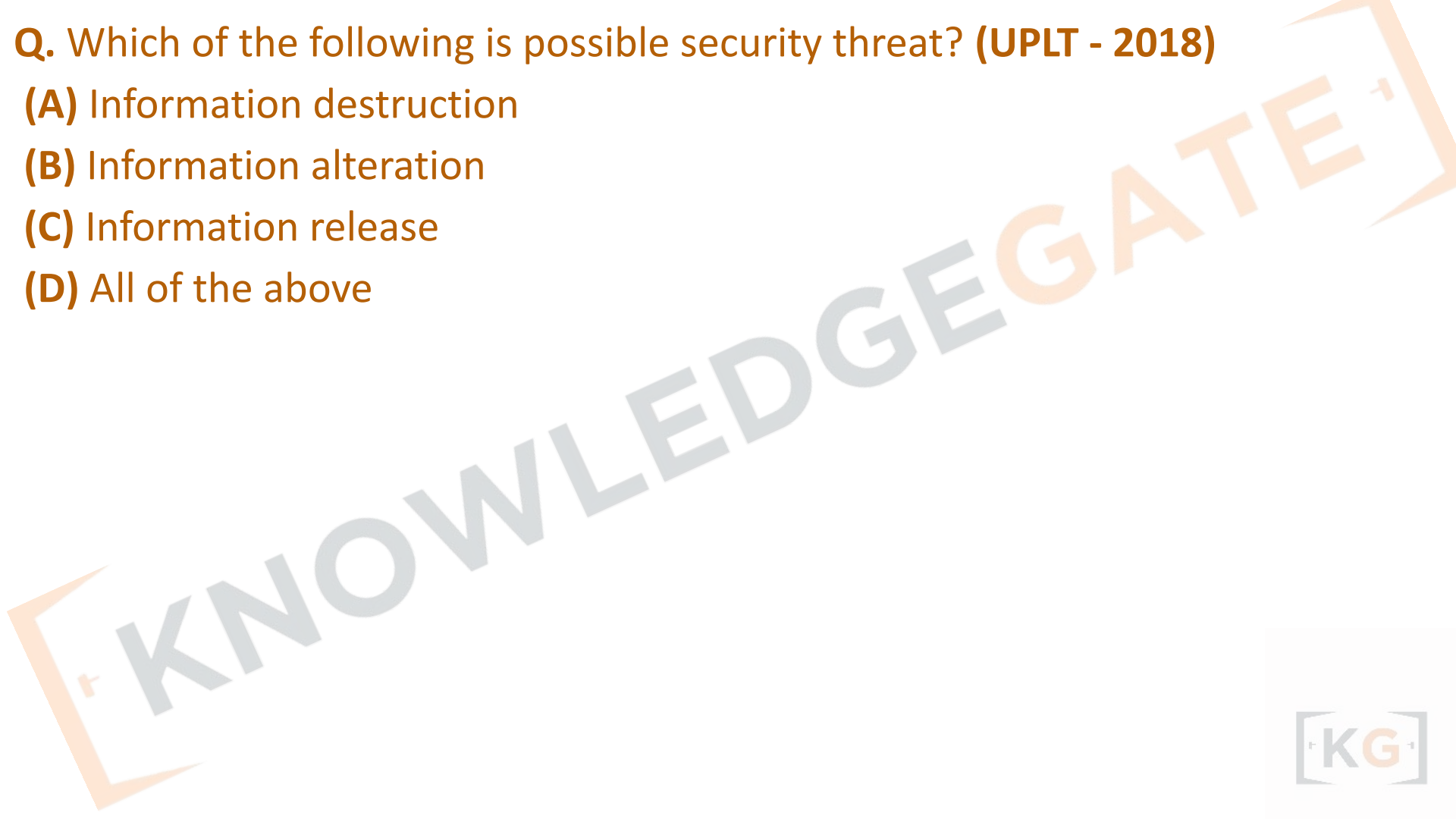


Q. _____ monitors user's activity on Internet and transmits that information in the background to someone else. (UPLT - 2018)

- (A) Malware**
- (B) Spyware**
- (C) Adware**
- (D) None of the above**

Q. A portion of the polymorphic virus, generally called a _____, creates a random encryption key to encrypt the remainder of the virus. **(UPLT - 2018)**

- (A)** Polymorphic engine
- (B)** Multiple engine
- (C)** Mutual engine
- (D)** Mutation engine



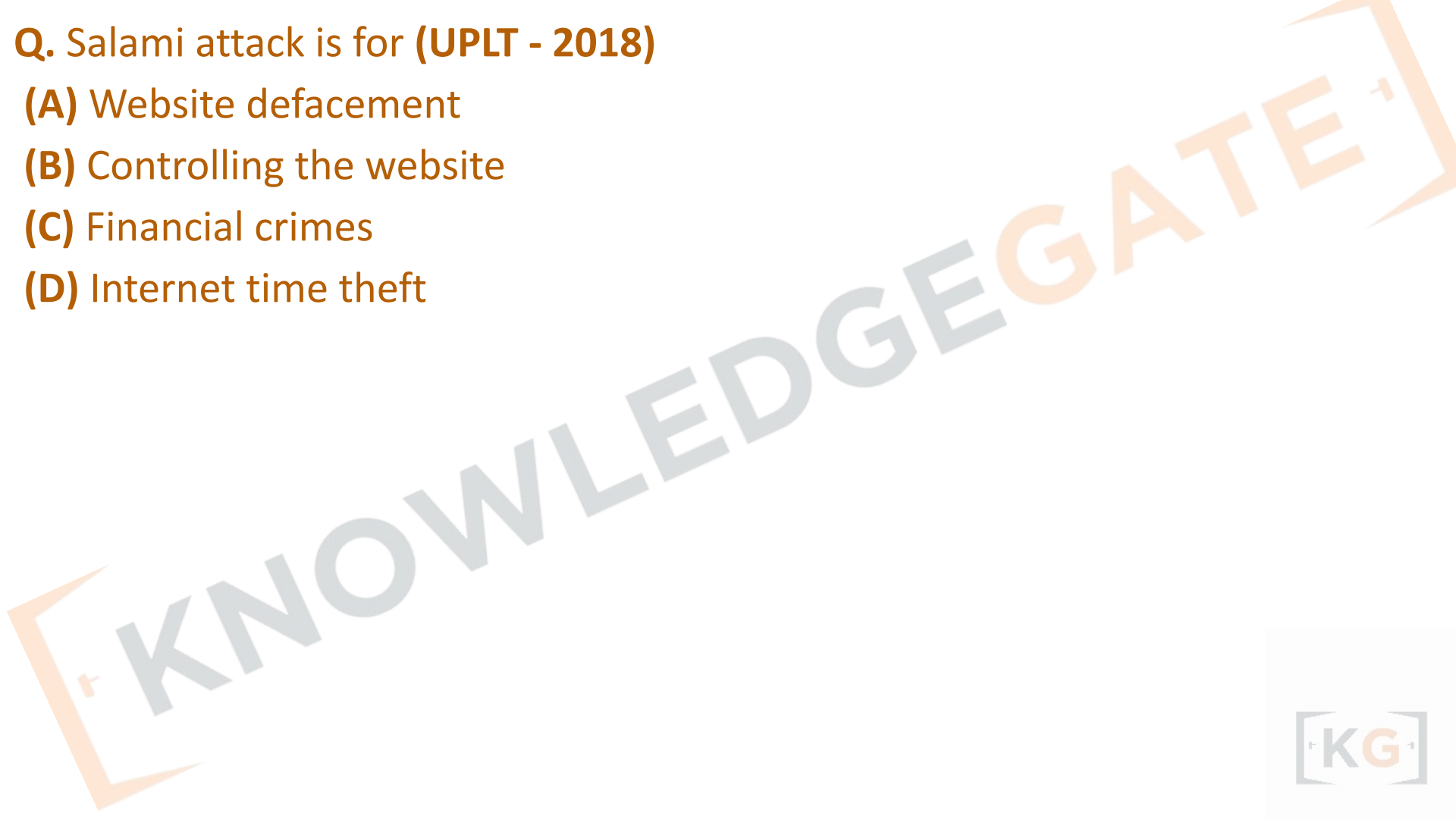
Q. Which of the following is possible security threat? (UPLT - 2018)

- (A)** Information destruction
- (B)** Information alteration
- (C)** Information release
- (D)** All of the above



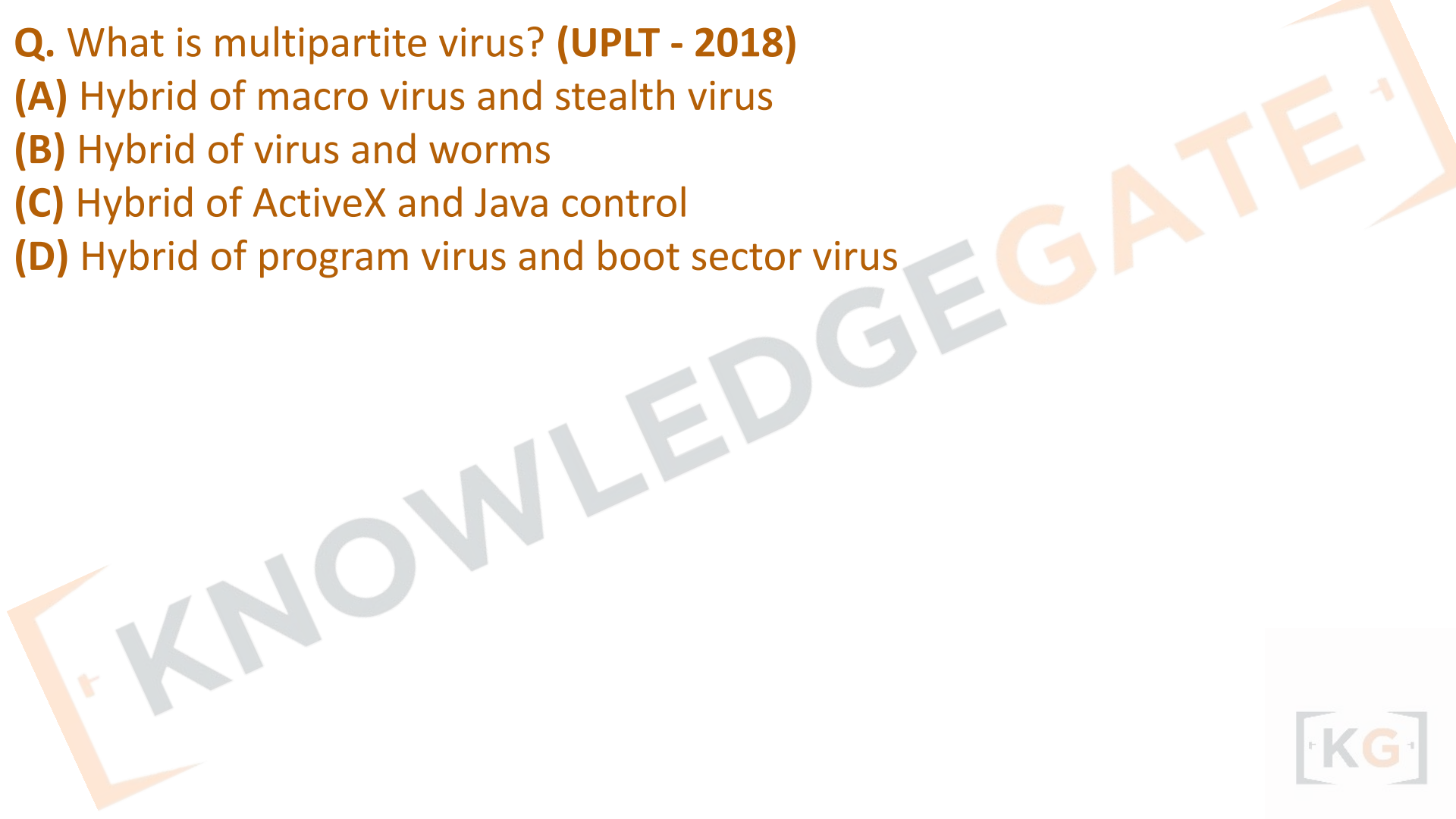
Q. Salami attack is for (UPLT - 2018)

- (A)** Website defacement
- (B)** Controlling the website
- (C)** Financial crimes
- (D)** Internet time theft



Q. What is multipartite virus? (UPLT - 2018)

- (A)** Hybrid of macro virus and stealth virus
- (B)** Hybrid of virus and worms
- (C)** Hybrid of ActiveX and Java control
- (D)** Hybrid of program virus and boot sector virus



Trojan-Horse programs:

- (a) Are legitimate programs that allow unauthorized access
- (b) Are hacker programs that do not show up on the system
- (c) Are really do not work usually
- (d) Are usually immediately discovered

TNPSC 2016 (Degree) P-II

Ans. (a)



A Trojan horse is a:

- (a) Special password that is built into the system firmware
- (b) Computer virus that spreads through email
- (c) Program that does something beyond its apparent purpose
- (d) Program that shut down if its initial condition are not met

TNPSC 2016 (Degree) P-II

Ans. (b)



Assessing-Damages

Assessing-Damages

In network security, **assessing damages** means **evaluating the impact, loss, and consequences** caused by a cyber attack, security breach, or system failure.

It helps an organization understand:

- What was affected
- How much data was lost
- What systems were compromised
- Financial and non-financial losses
- Recovery cost and future risk

Why Assess Damages?

- **Estimate total impact:** Helps understand how badly the attack affected the organization.
- **Calculate financial loss:** Determines money lost due to downtime, repair, and recovery efforts.
- **Identify legal responsibility:** Shows whether the organization is liable for data leaks or failures.
- **Plan recovery steps:** Guides how to restore systems and prevent future attacks.
- **Improve security policies:** Helps update weak areas in the security system.

Types of Damages

A. Direct Damages

- The immediate, measurable loss such as downtime, data deletion, or hardware failure.

B. Indirect Damages

- Loss that occurs later, like customer loss, reputation damage, or legal issues.

C. Tangible Damages

- Physical or monetary losses that can be easily calculated.

D. Intangible Damages

- Non-measurable losses such as reduced trust or damaged brand reputation.

Key Areas to Assess

1. Data Loss

- Measures how much data was stolen, altered, or destroyed during the attack.

2. System Impact

- Identifies which systems or services became unavailable due to the attack.

3. Financial Loss

- Calculates expenses for downtime, repair, legal fees, and regulatory penalties.

Key Areas to Assess

4. Operational Damage

- Evaluates how much business productivity and workflow were disrupted.

5. Reputational Damage

- Assesses how customer trust and public image were affected.

6. Legal & Compliance Damage

- Checks for any violations of security laws or industry regulations.

Steps to Assess Damages

Step 1: Identify the Incident

- Determine the type of attack and understand how it entered the system.

Step 2: Contain & Isolate

- Stop the attack quickly by disconnecting infected systems from the network.

Step 3: Evaluate Impact

- Identify which systems, data, and services were affected and how severely.

Steps to Assess Damages

Step 4: Collect Evidence

- Gather logs, files, and system records for forensic analysis and reporting.

Step 5: Quantify the Loss

- Convert system damage and data loss into measurable financial terms.

Step 6: Prepare Damage Report

- Create a formal document summarizing the attack and total estimated loss

Security in Mobile and Wireless Computing



Security in Mobile and Wireless Computing

Security in mobile and wireless computing involves protecting mobile devices and their networks from threats like malware, unauthorized access, and data interception. Key measures include encrypting data, using strong authentication, keeping wireless interfaces like Wi-Fi and Bluetooth disabled when not in use, and avoiding insecure public networks. By implementing these precautions, users can ensure the confidentiality, integrity, and availability of their data.



Authentication is the process of proving that people and organizations are who or what they claim to be. For wireless networks, this is often done at two layers:

- **The network layer:** The network requires the user to be authenticated before that person is granted access. This can be done implicitly, based on the device or modem being used.
- **The application layer:** Authentication is important at two levels — the client and the enterprise server. The simplest, and probably least secure, method of authentication is a username/password combination. More advanced methods include digital certificates or digital signatures.



Security in Mobile and Wireless Computing

Creating a Secure Environment

- For end-to-end security you have to consider the entire environment, including enterprise access, middle-tier components, and client applications.
- Objectives involved in creating a secure mobile environment:
 - **Authentication**
 - **Data Integrity**
 - **Confidentiality**
 - **Authorization**
 - **Nonrepudiation**

- **Data integrity** is assurance that the data has not been altered or corrupted in any way during the transmission from the sender to the receiver. This can be accomplished by using data encryption in combination with a cryptographic checksum or Message Authentication Code (MAC). When recipients receive the message, they compute the MAC and compare it with the MAC encoded in the message to see if the codes are the same.



- **Confidentiality** is one of the most important aspects of security, and certainly the most talked about. Confidentiality is about maintaining data privacy, making sure it cannot be viewed by unwanted parties. The most common way of providing confidentiality is by encrypting the data.



- **Authorization** is the process of determining the user's level of access—whether a user has the right to perform certain actions. Authorization is often closely tied to authentication. Once a user is authenticated, the system can determine what that party is permitted to do. Access control lists (ACLs) are often used to help determine this.



- **Non-repudiation** is about making parties accountable for transactions in which they have participated. It involves identifying the parties in such a way that they cannot at a later time deny their involvement in the transaction.



Security Threats

- Building a secure solution is difficult without awareness of the potential risks. The requirements for a secure environment, we will look at four common security threats: *spoofing*, *sniffing*, *tampering*, and *theft*.
- **Spoofing** is the attempt by a party to gain unauthorized access to an application or system by pretending to be someone he or she is not.
- If the spoofer gains access, he or she can then create fake responses to messages in an attempt to gain further knowledge and access to other parts of the system.



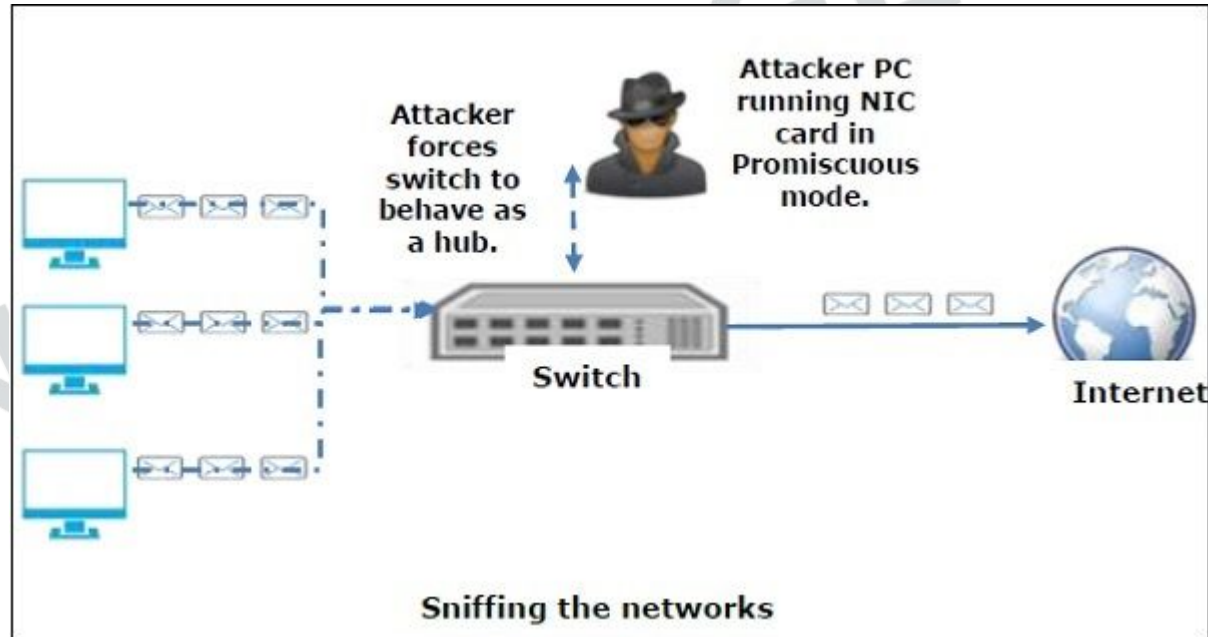
Security Threats

- If the spoofer gains access, he or she can then create fake responses to messages in an attempt to gain further knowledge and access to other parts of the system.
- Spoofer can make application users believe that they are communicating with a trusted source, such as their bank, when in reality they are communicating with an attacker machine.
- Unknowingly, users often provide additional information that is useful to the attacker to gain access to other parts and other users of the system.

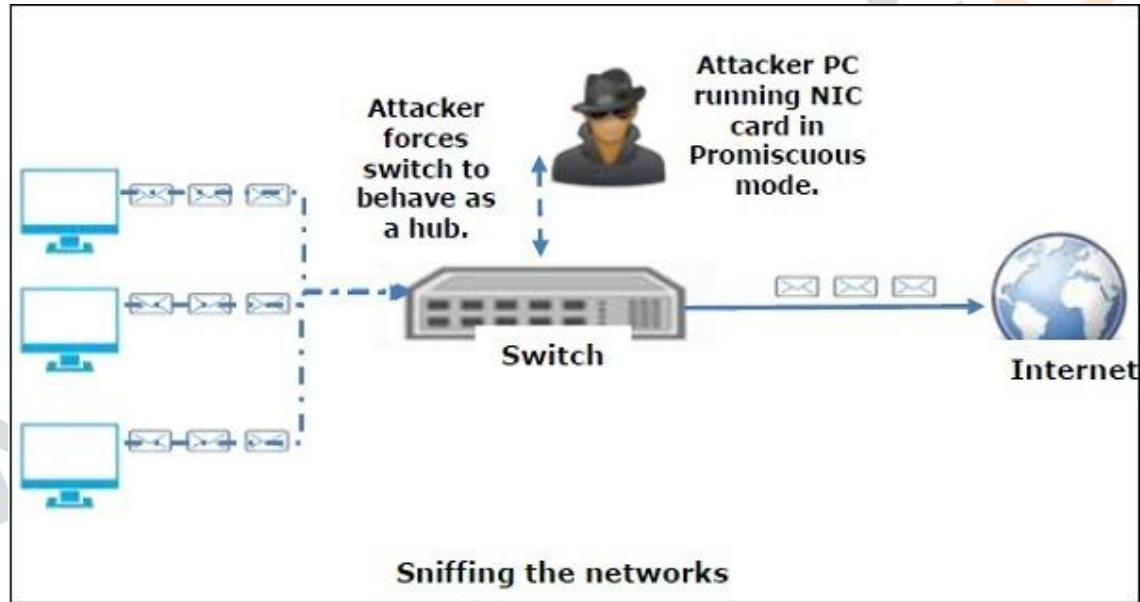


Sniffing is a technique used to monitor data flow on a network.

Sniffing can be used for proper purposes. But, it is more commonly associated with the unauthorized copying of network data. By "listening" to network data, unauthorized parties are able to obtain sensitive information that will allow them to do further damage to the application users, the enterprise systems, or both.



Sniffing is dangerous because it is both simple to do and difficult to detect. Moreover, sniffing tools are easy to obtain and configure. To combat the more sophisticated sniffing tools, data encryption is the best defense. Many wireless LAN users have discovered this.



Sniffing is used to perform_____ finger printing.

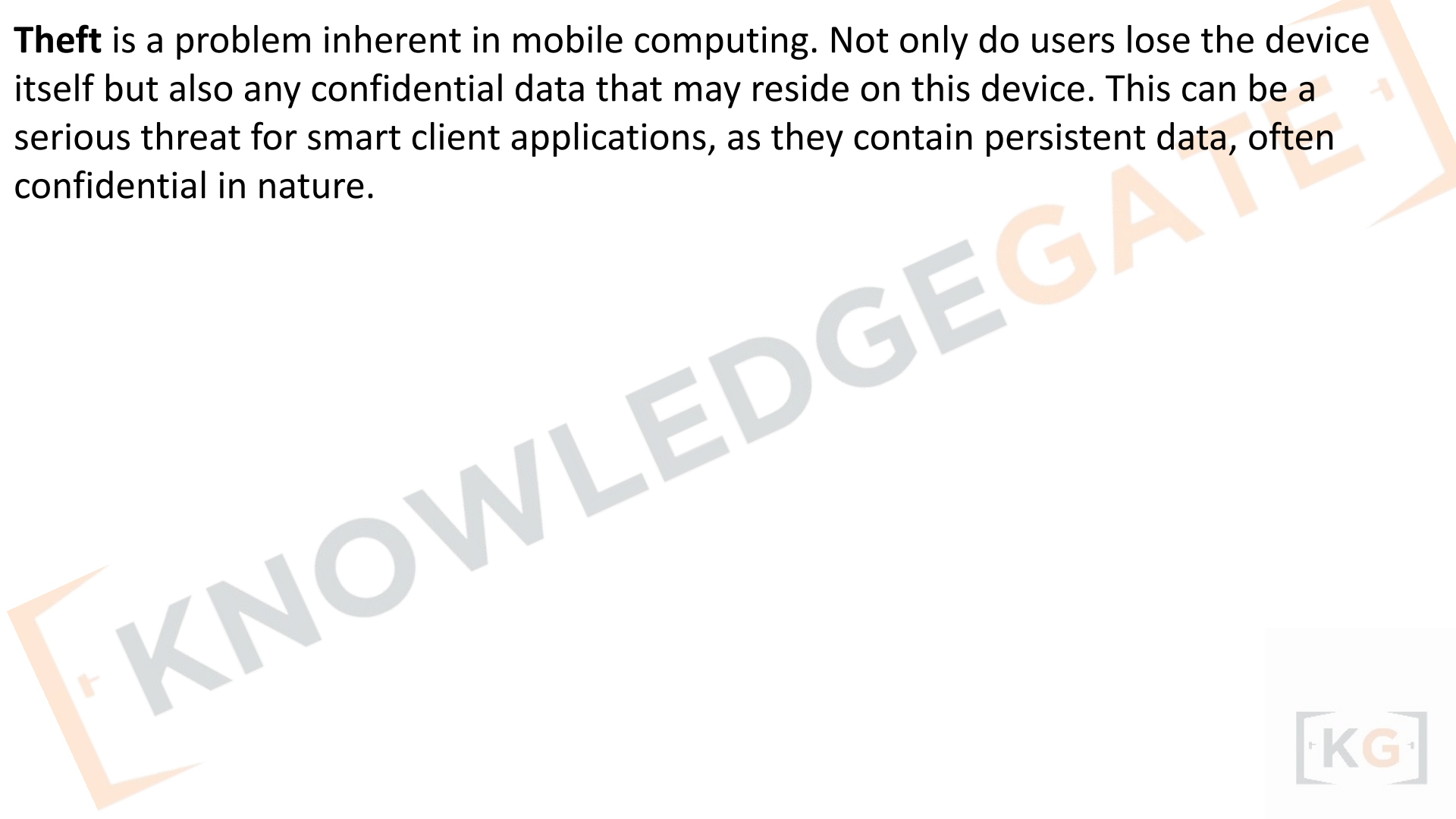
- (a) Passive stack
- (b) Active stack
- (c) Passive banner grabbing
- (d) Scanned

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Ans. (a)

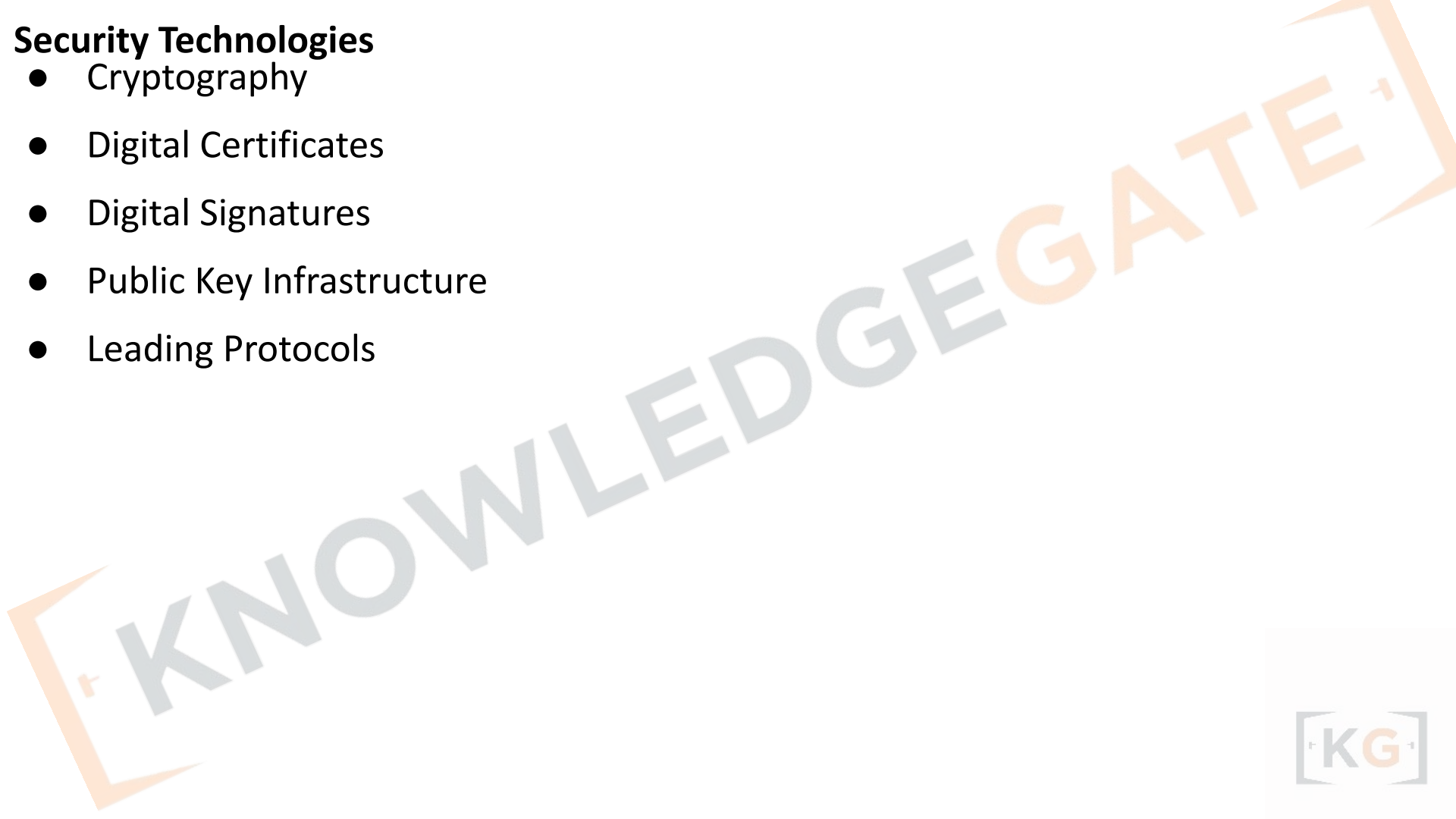


Theft is a problem inherent in mobile computing. Not only do users lose the device itself but also any confidential data that may reside on this device. This can be a serious threat for smart client applications, as they contain persistent data, often confidential in nature.



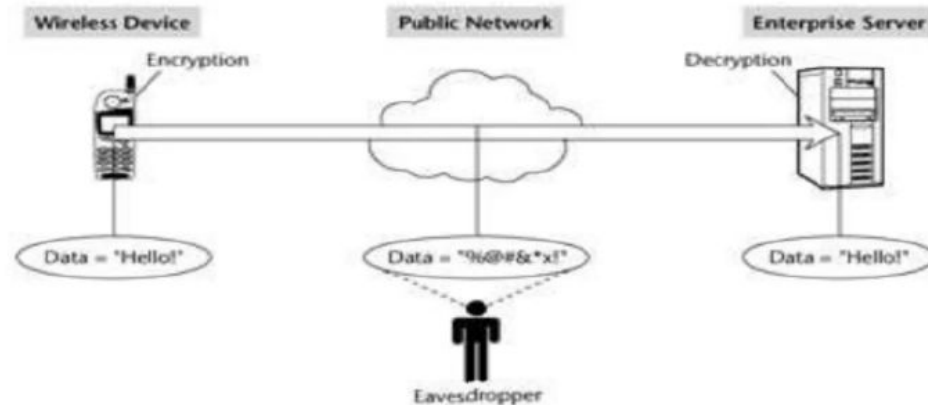
Security Technologies

- Cryptography
- Digital Certificates
- Digital Signatures
- Public Key Infrastructure
- Leading Protocols

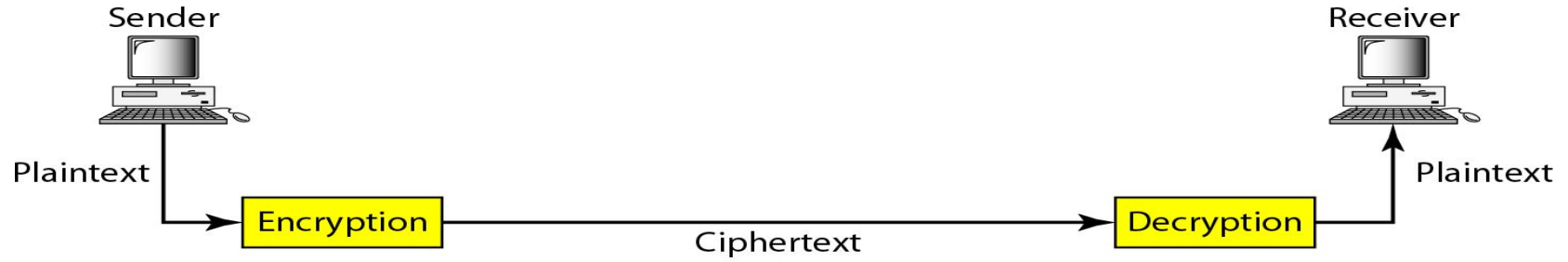


Cryptography

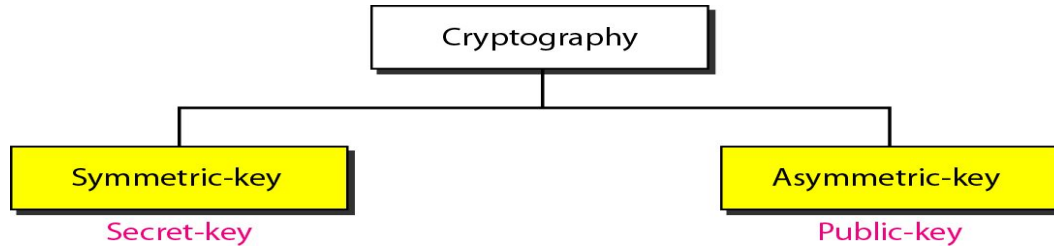
- The basic objective of cryptography is to allow two parties to communicate over an insecure channel without a third party being able to understand.
- **Algorithms and Protocols** – Cryptography works on many levels.
 - At the lowest level are cryptographic algorithms.
 - The protocol describes the complete process of executing a cryptographic activity, including explicit information on how to handle any contingency that might arise.



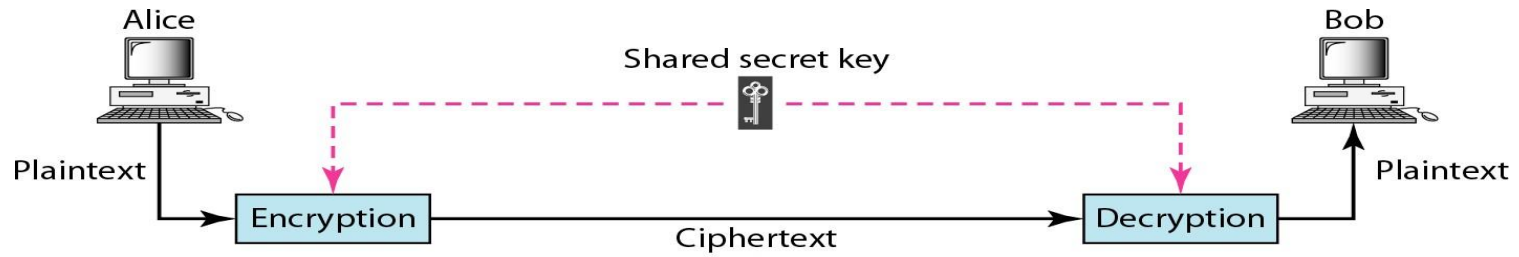
Cryptography components



Categories of cryptography



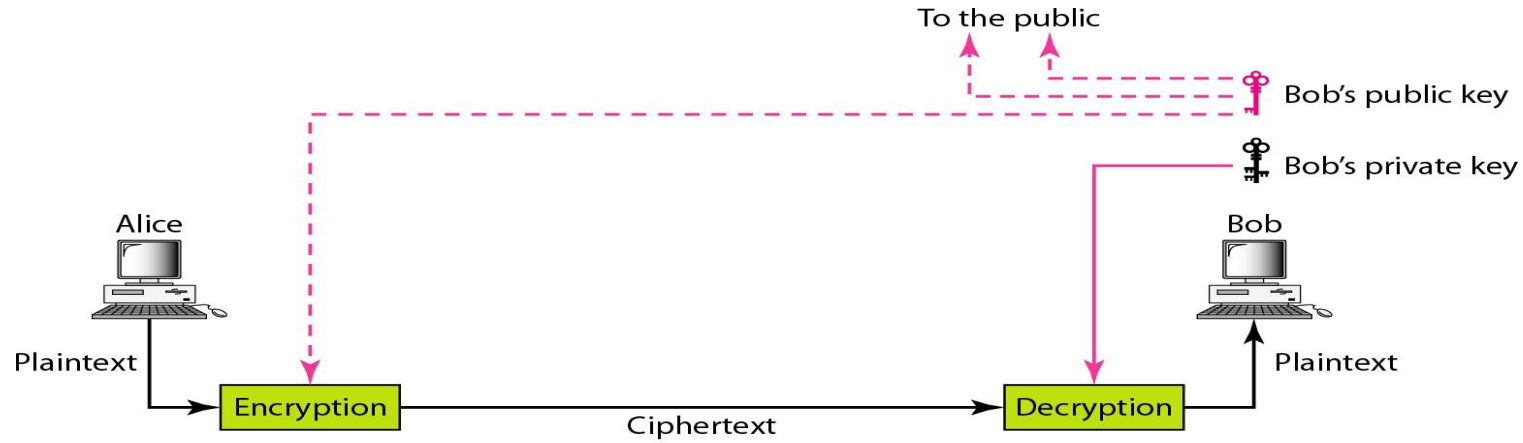
Symmetric-key cryptography





In symmetric-key cryptography, the same key is used by the
sender
(for encryption)
and the receiver (for decryption).
The key is shared.

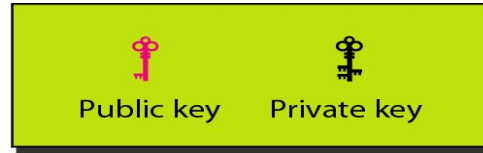
Asymmetric-key cryptography



Keys used in cryptography

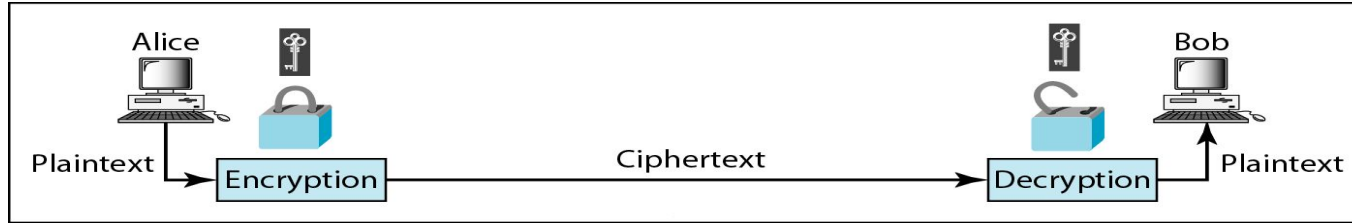


Symmetric-key cryptography

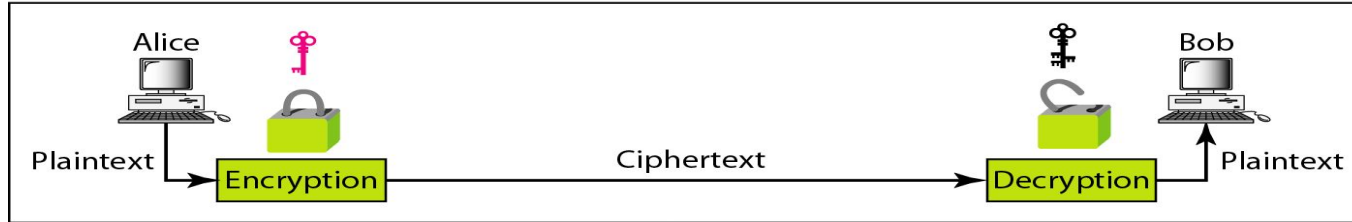


Asymmetric-key cryptography

Comparison between two categories of cryptography



a. Symmetric-key cryptography



b. Asymmetric-key cryptography

The art and science of concealing the messages to introduce secrecy in information security is recognized as :

- (a) Scientometrics
- (b) Cryptography
- (c) Biography
- (d) Information forensics

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Ans. (b)



The encryption process where same keys are used for encrypting and decrypting the information is known as _____ encryption.

- (a) Symmetric Key
- b) Asymmetric Key
- (c) Linear
- (d) Tertiary Key

Digital Certificates

- Digital certificates provide a way to guarantee that a public key belongs to the party it represents.
- For this to be successful, the certificate itself also has to be verified to ensure that it represents the claimed entity (a person or organization). This is accomplished using a trusted third party called a certificate authority (CA).
- Digital certificates typically contain the following:
 - The name of the holder, as well as other information that uniquely identifies the holder.
 - The holder's public key.
 - The name of the CA that issued the certificate.
 - The lifetime that the certificate is valid for (usually a start and end date).
 - A digital signature from the CA.

Leading Protocols

- Secure Sockets Layer (SSL)
- Transport Layer Security (TLS)
- Wireless Transport Layer Security (WTLS)
- IP Security (IPSec)

Secure Sockets Layer

- SSL is the dominant security protocol being used on the Internet today.
- It was developed by Netscape to provide secure and private Internet sessions, typically on top of HTTP, FTP, and other relevant protocols.
- SSL uses a combination of symmetric and asymmetric algorithms to maximize performance.

Transport Layer Security

- Transport Layer Security (TLS) is the next generation of SSL.
- It consists of two layers. The lower layer is the TLS Record protocol, which is layered on top of a reliable transport protocol such as TCP.
- The two main features of the Record protocol are private and reliable connections.
- The higher level is the TLS Handshake protocol. This layer provides connection security that authenticates using asymmetric cryptography, negotiates a secret key, and provides a reliable negotiation.

Wireless Transport Layer Security

- WTLS is the security layer defined in the WAP specification.
- It operates above the Transport Protocol Layer, making it suitable for a variety of underlying wireless protocols.
- It is similar to TLS, but is optimized for low-bandwidth networks with high latency.
- It also adds new features such as datagram support, optimized handshakes, and key refreshing.

IP Security

- IPsec is different from the other protocols in that it does not operate on the application layer.
- SSL, TLS, and WTLS are aimed at providing secure communications over an inherently insecure network, whereas IPsec is aimed at making the Internet itself secure.
- It does this by providing authentication, integrity, and privacy services at the IP datagram layer.
- IPsec will become a more prominent solution when mobile devices start to support IP6, which includes IPsec as part of the standard.

Other Security Measures

- Firewalls
- Virtual Private Networks
- Two-Factor Authentication
- Biometrics

Firewalls

- Firewalls are the most common form of security implemented within organizations. They establish a network perimeter between what is public and what is private. A firewall is a set of software programs, usually located on a separate gateway server that restricts access to the private network resources from users on other networks.
- At a lower level, a firewall will examine each packet of network data to determine whether it should be forwarded to its destination.
- For mobile devices that have always-on connections, a personal firewall may also be useful. A personal firewall will prevent not only suspicious connections but also inappropriate content.

Virtual Private Networks

- A VPN allows a company to turn a public network into a private network.
- This technology allows remote workers to communicate with the corporate network in a secure fashion.
- Before VPN technology became available, dedicated leased lines were required to achieve the same result.
- VPNs present an additional benefit over leased lines, by providing secure access from many locations—essentially anywhere an Internet connection is available.
- Mobile VPNs for devices on public networks are still in the early phases of adoption.

Which of the following makes (s) filtering decisions based on application payload?

- (a) Packet filter
- (b) deep inspection firewall
- (c) reverse proxy
- (d) stateful packet inspection firewall

Ans. (b)

TRB Poly. Lect. 2017



Which of the following is not a factor in choosing a firewall?

- (a) The applications that your network uses
- (b) Information in your security policy
- (c) How much traffic will pass through the firewall
- (d) Security audit measures in place

Ans. (b)

Punjab PSC Network Engineer 21.06.2014



Which of the following types of firewall works in the network layer of the OSI Model?

- (a) Packet filters
- (b) Application gateways
- (c) Circuit level gateways
- (d) Both application level and circuit level gateways

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Ans. (a) :



Two-Factor Authentication

- For some purposes, usually dealing with financial transactions, strong authentication is required.
- This involves using a two-factor approach, where users have to apply two factors to authenticate themselves.
- One factor is usually something the user knows, such as a PIN number; the other is something the user has, such as a token card to generate a one-time password.
- Smart client applications inherently provide a form of two-factor authentication: First you must have the device to access the application; second, you must authenticate yourself to gain access to the application, as well as to any back-end system to which it connects.

Biometrics

- Even with the increased security of two-factor authentication, unauthorized users can compromise the system, for example, by obtaining the PIN code and token card, thereby gaining access to the enterprise system.
- Stronger form of authentication: biometric authentication.
- Biometrics provides a wide range of techniques for authenticating an individual based on his or her unique physical characteristics. Such techniques include fingerprint identification, face recognition, voice recognition, or iris and retina scanning.
- Using biometric techniques, you can ensure that the identification token is indeed unique.

Security Policy

- Most important security measure is the adoption of a corporate security policy. Such a policy will outline all aspects of a corporation's security measures, including both technology and the use and disclosure of confidential information within the enterprise. Even if a corporation has implemented a very strong technical security solution, the overall system will still be insecure if its users do not follow corporate security guidelines. Remember, intruders will always attack the weakest link in a system. Unfortunately, this link is often the users themselves.
- Sometimes, very simple measures will dramatically increase overall security. For example, many PDA users do not lock the operating system when it is not in use. If the device is lost, nothing prevents another user from accessing the applications and corresponding data on the device.

E-Governance and EDI

E-Governance

Electronic Governance or E-Governance is the application of Information and Communication Technology (ICT) for providing government services, interchange of statistics, communication proceedings, and integration of various independent systems and services. Through the means of e-governance, government services are made available to citizens in a suitable, systematic, and transparent mode. The three main selected groups that can be discriminated against in governance concepts are government, common people, and business groups.

E-governance is the best utilization of information and communication technologies to mutate and upgrade the coherence, productivity, efficacy, transparency, and liability of informational and transnational interchanges within government, between government agencies at different levels, citizens & businesses. It also gives authorization to citizens through access and use of information. Generally, E-governance uses information and communication technologies at various levels of the government and the public sector to enhance governance.

Theoretical studies state that E-Governance is the procedure of change of the correlation of government with its ingredients, the citizens, the businesses, and its own organs, through the use of tools of information and communication technology. The UNESCO states that E-governance is the public sector's use of information and communication automation in order to upgrade information and service delivery, stimulating resident involvement in the decision-making process and making government more liable, unambiguous and productive.

Elements of E-Governance:

Basic elements of e-governance are:

1. Government
2. Citizens
3. Investors/Businesses

Types of E-Governance:

E-governance is of 4 types:

1. Government-to-Citizen (G2C): The Government-to-citizen mentions the government services that are acquired by the familiar people. Most of the government services come under G2C. Similarly, the primary aim of Government-to-citizen is to supply facilities to the citizens. It also helps ordinary people to minimize the time and cost to carry out a transaction. A citizen can retrieve the facilities anytime from anywhere. Similarly, spending the administrative fee online is also possible due to G2C. The facility of Government-to-Citizen allows the ordinary citizen to outclass time limitations. It also focuses on geographic land barriers.

Types of E-Governance:

2. Government-to-business (G2B): Government-to-business is the interchange of services between Government and Business firms. It is productive for both government and business firms. G2B provides access to pertinent forms needed to observe. It also contains many services interchanged between business sectors and government.

Similarly, Government-to-business provides timely business information. A business organization can have easy and easy online access to government agencies. G2B plays an important role in business development. It upgrades the efficiency and quality of communication and transparency of government projects.

3. Government-to-Government (G2G): The Government-to-Government mentions the interaction between different government departments, firms, and agencies. This increases the efficiency of government processes. In G2G, government agencies can share the same database using online communication. The government departments can work together. This service can increase international discretion and relations. G2G services can be at the local level or at the international level. It can convey to both global government and local government. It also provides a safe and secure inter-relationship between domestic and foreign governments. G2G builds a universal database for all members to upgrade service.

4. Government-to-Employee (G2E): The Government-to-Employee is the internal part of G2G section. It aims to bring employees together and improvise knowledge sharing. It provides online facilities to the employees. Similarly, applying for leave, reviewing salary payment record and checking the balance of holiday. The G2E sector yields human resource training and development. So, G2E is also the correlation between employees and government institutions.

Advantages of E-Governance:

The supreme goal of e-governance is to be able to provide an increased portfolio of public services to citizens in a systematic and cost effective way. It allows for government transparency because it allows the public to be informed about what the government is working on as well as the policies they are trying to implement.

The main advantage while executing electronic government will be to enhance the efficiency of the current system.

Another advantage is that it increases transparency in the administration, reduces costs, increases revenue growth, and also improves relationships between the public and the civic authorities.

Disadvantages of E-Governance:

The main disadvantage regarding e-governance is the absence of fairness in public access to the internet, of trustworthy information on the web, and disguised agendas of government groups that could have an impact and could bias public opinions.

The primary objective of e-Governance is:

- A) Increase paperwork
- B) Deliver government services electronically
- C) Replace government employees
- D) Increase taxation

✓ **Answer: B**

Explanation:

E-Governance aims to **digitize government services** for transparency, speed, and citizen convenience.

Which of the following is NOT a model of e-Governance?

- A) G2C
- B) G2B
- C) G2E
- D) B2B

✓ **Answer: D**

Explanation:

B2B is a business-to-business commercial model, **not** an e-Governance model.

E-Governance includes **G2C, G2B, G2E, and G2G**.



Which of the following is an example of a G2C service?

- A) Filing income tax return
- B) Vendor registration
- C) Inter-department file movement
- D) Payroll management

✓ **Answer: A**

Explanation:

Filing taxes, Aadhaar services, voting registration etc. are **Government-to-Citizen (G2C)** services.



Electronic Data Interchange (EDI)

Electronic Data Interchange (EDI) is a computer-to-computer exchange of business documents in a standard electronic format between two or more trading partners. It enables companies to exchange information electronically in a structured format, eliminating the need for manual data entry and reducing the cost and time associated with paper-based transactions.

EDI was first introduced in the 1960s as a way for companies to exchange business documents electronically. Over time, the standardization of EDI formats and protocols has enabled businesses to integrate their internal systems with those of their trading partners, improving efficiency and reducing errors.

EDI transactions can include purchase orders, invoices, shipping notices, and other business documents. The EDI standard defines the format and content of these documents, ensuring that they are easily interpreted by both the sender and the receiver.

EDI has become an important part of many businesses, particularly those in the supply chain and logistics industries. It allows for faster and more accurate processing of transactions, leading to improved customer satisfaction and increased profits.

Imagine writing a letter to your friend while communicating every time, Can not imagine right? Since today humans live in an era where they can very easily communicate through the internet. Now, imagine the same case with businesses, where communication and exchange of very important documents are constantly required, doing this the old way, it will take forever for the messages to reach the other party, but also the documents will pile up as there is a lot of information that is needed to be stored and kept. It is a tedious and cumbersome process indeed, this is where EDI plays its role.

Electronic Data Exchange is the direct exchange of data and important business documents through the Internet and in a very professional manner. Two different companies sitting at the extreme corners of the world can very easily interchange information or documents (like sales orders, shipping notices, invoices, etc) with the help of EDI.

KNOWLEDGEGE



EDI Documents:

The most common documents exchanged via EDI are:

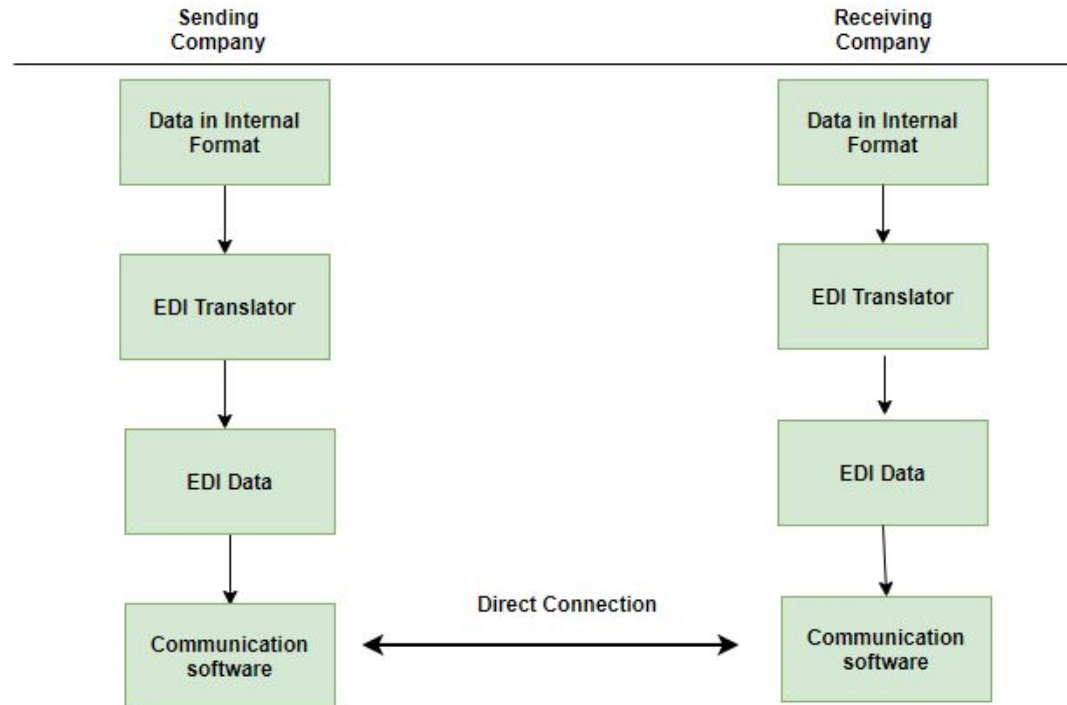
- Invoices
- Purchase Orders
- Financial Information letters
- Transaction Bills
- Shipping requests and notifications
- Acknowledgment and feedback
- Transcripts
- Claims
- Business Correspondence letters

EDI Users:

- Central and state government agencies
- Industry
- Banking
- Retailing
- Manufacturing
- Insurance
- Healthcare
- Automotive
- Electronics
- Grocery
- Transportation

How EDI works?

The data or the information that one company sends the other first gets prepared to be sent, then the information/document is translated into EDI format. The document is then connected and transmitted to the other business, the connection is direct and point to point.



Uses of EDI :

EDI is widely used in various industries for exchanging business documents electronically. Some of the common uses of EDI are:

- **Order Processing:** EDI allows companies to exchange purchase orders and sales orders electronically, eliminating the need for manual data entry and reducing errors.
- **Invoicing:** EDI can be used to exchange invoices electronically, reducing the time and cost associated with paper-based invoicing.
- **Shipping and Receiving:** EDI can be used to exchange shipping notices and receiving documents, enabling companies to track the movement of goods in real-time.

Uses of EDI :

- **Inventory Management:** EDI can be used to exchange inventory information, enabling companies to manage their inventory levels more effectively.
- **Supply Chain Management:** EDI is used extensively in the supply chain management process, enabling companies to exchange information with their suppliers, distributors, and customers.

- **Healthcare:** EDI is used in the healthcare industry to exchange patient data, claims, and other healthcare-related information between healthcare providers, insurance companies, and government agencies.
- **Financial Transactions:** EDI can be used to exchange financial transactions such as payment advice and remittance advice, reducing the time and cost associated with manual payment processing.

Advantages of EDI:

There are several advantages to Electronic Data Interchange:

- **The paper usage reduced:** The expense of storing, printing, recycling, reduces up to the maximum amount due to the EDI.
- **Improved quality of Data:** The data entry errors are reduced due to EDI.
- **Speed Increases:** The best advantage is the increase in the speed of the data interchange. With everything going online, the speed of the information transfer increases exponentially.
- **Security:** By following the Protocols and the standard rules, the security of all the important documents is always secure and safe.
- **Information accuracy:** Since the information exchanged is based on standards agreed by the sender and receiver both, the correct information is always transferred regardless of where they belong to.
- **Less Cost:** With very less errors, fast response time, every thing becoming automated, and no use of paper, the cost automatically reduces.

Disadvantages of EDI:

- The initial setup of the EDI is very Time-consuming.
- EDI standards keep on changing after some amount of time.
- A very systematic and proper back up is required as the entire data relies on EDI.
- The setup and maintenance of the EDI is very Expensive.

EDI is used mainly for:

- A) Sending emails
- B) Electronic exchange of business documents
- C) Video conferencing
- D) Web browsing

✓ **Answer: B**

Explanation:

EDI enables **computer-to-computer exchange** of structured business documents.



Which of the following is NOT an advantage of EDI?

- A) Faster document processing
- B) Reduced errors
- C) Increased transaction cost
- D) Improved accuracy

✓ **Answer: C**

Explanation:

EDI **reduces** transaction cost by eliminating paperwork and manual processing.



Which protocol is commonly used in EDI for data transmission?

- A) SMTP
- B) FTP
- C) HTTP
- D) POP3

✓ **Answer: B**

Explanation:

FTP is widely used for transferring EDI documents securely between trading partners.



What is E-Commerce?

E-Commerce stands for Electronic commerce, which means buying or selling goods through the Internet. The biggest advantage of E-Commerce in this era is Time Savings, not only that as a customer, one major advantage is that the customer receives a lot of discounts on the products they want to buy.

In terms of business, a businessman not only can expand the market all over the country but also all around the world. Businesses also do not need to put too much effort into Branding.

One major thing that comes to play its role in E-Commerce is communicating professionally. Let's learn about this in further detail,

Types of E-commerce model:

There are types of e-commerce models that can describe almost every transaction that takes place between consumers and businesses.

1. **Business to Consumer (B2C):** When a good or service is sold to an individual consumer by a business, e.g., we buy a pair of shoes from an online retailer.
2. **Business to Business (B2B):** When a good or service is sold by a business to another business, e.g., a software-as-a-service is sold by a business for other businesses to use.
3. **Consumer to Consumer (C2C):** When a good or service is sold by a consumer to another consumer, e.g., we sell our old furniture on eBay to another consumer.

Types of E-commerce model:

3.. Consumer to Business (C2B): When a consumer's own products or services is sold to a business or organization, e.g., an authority offers exposure to their online audience in exchange for a fee or a photographer licenses their photo for a business to use.

4. Business-to-government (B2G): In this model, businesses sell products or services to government agencies through online portals or marketplaces.

5. Government-to-business (G2B): In this model, government agencies sell products or services to businesses through online portals or marketplaces.

6. Government-to-consumer (G2C): In this model, government agencies provide products or services directly to consumers through online portals or marketplaces.

Two main types of Internet-based B2B commerce are

- (a) net marketplaces and private industrial networks
- (b) EDI and collaborative commerce
- (c) net marketplaces and collaborative commerce
- (d) EDI and private industrial networks

By electronic commerce we mean

- (a) Commerce of electronic goods
- (b) Commerce which depends on electronics
- (c) Commerce which is based on the use of internet
- (d) Commerce which is based on transactions using computers connected by telecommunication network

RPSC VPITI 2018 (IT) Ans. (d)

Ans. (d)



Snapdeal.com comes under which of the following e-commerce categories?

- (a) B2B
- (b) B2C
- (c) C2B
- (d) B2G
- (e) C2C

E-commerce infrastructure involves

- (a) web servers
- (b) e-catalogs
- (c) networks
- (d) All the given options

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Ans. (d)



Security Threats To E-commerce

E-Commerce security requirements can be studied by examining the overall process, beginning with the consumer and ending with the commerce server.

Considering each logical link in the commerce chain, the assets that must be protected to ensure secure e-commerce include client computers, the messages travelling on the communication channel, and the web and commerce servers – including any hardware attached to the servers.

Top 10 E-commerce Security Threats

1. Financial frauds

Ever since the first online businesses entered the world of the internet, financial fraudsters have been giving businesses a headache. There are various kinds of financial frauds prevalent in the e-commerce industry, but we are going to discuss the two most common of them.

a. Credit Card Fraud

It happens when a cybercriminal uses stolen credit card data to buy products on your e-commerce store. Usually, in such cases, the shipping and billing addresses vary. You can detect and curb such activities on your store by installing an [AVS](#) – Address Verification System.

Another form of credit card fraud is when the fraudster steals your personal details and identity to enable them to get a new credit card.

b. Fake Return & Refund Fraud

The bad players perform unauthorized transactions and clear the trail, causing businesses great losses. Some hackers also engage in refund frauds, where they file fake requests for returns.

2. Phishing

Several e-commerce shops have received reports of their customers receiving messages or emails from hackers masquerading to be the legitimate store owners. Such fraudsters present fake copies of your website pages or another reputable website to trick the users into believing them. For example, see this image below. A seemingly harmless and authentic email from PayPal asking to provide details.

3. Spamming

Some bad players can send infected links via email or social media inboxes. They can also leave these links in their comments or messages on blog posts and contact forms. Once you click on such links, they will direct you to their spam websites, where you may end up being a victim.

Mass-mailed malware infection can quickly morph into a much more serious problem says Brian Krebs, data security expert.

Apart from lowering your website security, spamming also reduces its speed and severely affects performance.

4. Malware

Hackers may design a malicious software and install on your IT and computer systems without your knowledge. These malicious programs include spyware, viruses, trojan, and ransomware. The systems of your customers, admins, and other users might have Trojan Horses downloaded on them. These programs can easily swipe any sensitive data that might be present on the infected systems and may also infect your website.

5. Exploitation of Known Vulnerabilities

Attackers are on the lookout for certain vulnerabilities that might be existing in your e-commerce store.

Often an e-commerce store is vulnerable to SQL injection (SQLi) and Cross-site Scripting (XSS).

Let's take a quick look at these vulnerabilities:

a. SQL Injection

It is a malicious technique where a hacker attacks your query submission forms to be able to access your backend database. They corrupt your database with an infectious code, collect data, and later wipe out the trail.

b. Cross-Site Scripting (XSS)

The attackers can plant a malicious JavaScript snippet on your e-commerce store to target your online visitors and customers. Such codes can access your customers' cookies and compute. You can implement the Content Security Policy (CSP) to

6. Bots : Some attackers develop special bots that can scrape your website to get information about inventory and prices. Such hackers, usually your competitors, can then use the data to lower or modify the prices in their websites in an attempt to lower your sales and revenue.

7. Brute force: The online environment also has players who can use brute force to attack your admin panel and crack your password. These fraudulent programs connect to your website and try out thousands of combinations in an attempt to obtain your site's passwords. Always ensure to use strong, complex passwords that are hard to guess. Additionally, always change your passwords frequently.

8. Man in The Middle (MITM)

A hacker may listen in on the communication taking place between your e-commerce store and a user. Walgreens Pharmacy Store experienced such an incident. If the user is connected to a vulnerable Wi-Fi or network, such attackers can take advantage of that.

9. e-Skimming

E-skimming involves infecting a website's checkout pages with malicious software. The intention is to steal the clients' personal and payment details.

Are you an e-commerce business person? Don't downplay the seriousness of these e-commerce security threats.

10. Malicious codes

Computer viruses, worms and trojan horses are examples of malicious code. A trojan horse is a program which performs a useful function, but performs an unexpected action as well. Virus is a code segment which replicates by attaching copies to existing executables. A DEPT OF CSE & IT VSSUT, Burla worm is a program which replicates itself and causes execution of the new copy. These can create havoc on the client side.

Concepts in Electronics payment systems, E-Cash, Credit/Debit Cards

What is an Electronic Payment System?

Electronic Payment System allows people to make online payments for their purchases of goods and services without the physical transfer of cash and cheques, irrespective of time and location. The key components of this payment system are the payers and payees, financial institutions, electronic devices, communication networks, payment gateways, and mobile payment apps. As the global economy continues to evolve, the dependency on physical modes of payment is gradually giving way to digital alternatives that offer speed, convenience, and efficiency. These systems facilitate a diverse range of financial activities, from online purchases and bill payments to person-to-person transfers.

Types of Electronic Payment System

India, being the fastest-growing economy and a developing nation, has witnessed significant growth in various types of Electronic Payment Systems, driven by technological advancements and efforts to promote a cashless economy. The prominent types of Electronic Payment Systems in India range

from the Unified Payments Interface (UPI) to Debit and Credit cards. Listed below are the types of Electronic Payment Systems:

1. Unified Payments Interface (UPI): UPI has become a widely adopted and popular electronic payment system in India. It enables users to link multiple bank accounts to a single mobile application, allowing seamless and instant fund transfers between individuals and merchants.

Types of Electronic Payment System

2. Mobile Wallets: Mobile Wallet services like Paytm, PhonePe, and Google Pay have gained widespread acceptance. Users can load money into these digital wallets and use the balance for various transactions, including mobile recharge, bill payments, and online shopping.

3. Debit and Credit Cards: Debit and Credit card usage is prevalent in India, with various banks issuing these cards for electronic transactions. Cards are commonly used for Point-of-Sale (POS) transactions, online purchases, and cash withdrawals from ATMs.

4. Immediate Payment Service (IMPS): IMPS enables instant interbank electronic fund transfers through mobile phones, internet banking, or ATMs. It is particularly useful for peer-to-peer transactions and small-value payments.

5. National Electronic Funds Transfer (NEFT): NEFT is a nationwide electronic payment system that facilitates one-to-one funds transfer between bank accounts. It operates on a deferred settlement basis and is widely used for both individual and corporate transactions.

6. Real-Time Gross Settlement (RTGS): RTGS is another electronic fund transfer system that allows real-time settlement of large-value transactions. It is typically used for high-value interbank transfers.

7. Prepaid Instruments: Prepaid Instruments, including prepaid cards and gift cards, provide users with a convenient way to make electronic payments with a pre-loaded amount.

Advantages of Electronic Payment System

- **24/7 Accessibility:** Electronic Payments can be made at any time, providing round-the-clock access to financial transactions.
- **Global Accessibility:** Users can make payments and transfer funds globally without being restricted by geographical boundaries.
- **Instant Transactions:** Electronic Payments are processed quickly, allowing for near-instantaneous transfer of funds between accounts.
- **Faster Settlement:** Compared to traditional payment methods, electronic transactions often result in faster settlement times.
- **Record-Keeping and Tracking:** Electronic Payment Systems facilitate easy record-keeping for both businesses and individuals.
- **Encryption and Authentication:** Electronic Payment Systems employ robust encryption and authentication protocols to secure transactions and protect sensitive information.

Disadvantages of Electronic Payment System

- **Security Concerns:** Electronic Payment Systems are susceptible to security breaches, including hacking, phishing, and identity theft.
- **Technical Issues:** Electronic Payment Systems rely on technology, and technical glitches or system failures can disrupt transactions.
- **Fraud Risk:** Despite security measures, Electronic Payment Systems are not immune to fraud. Unauthorized transactions, stolen credentials, or fraudulent activities can occur, leading to financial losses for individuals and businesses.
- **Privacy Concerns:** Users may be concerned about the collection and storage of personal information by electronic payment providers.
- **Transaction Fees:** Some electronic payment systems impose transaction fees, which can add up over time.

Digital Token Based Electronic Payment System

A digital token-based electronic payment system is a financial technology (FinTech) solution that facilitates secure and efficient transactions using digital tokens rather than traditional currencies like cash or credit cards. These systems leverage blockchain technology or other cryptographic methods to create and manage digital tokens, enabling a variety of payment scenarios. Here are key components and features of a digital token-based electronic payment system:

Digital Token Based Electronic Payment System

1. Digital Tokens: Digital tokens are units of value that represent real or virtual assets. These tokens are created, stored, and transacted electronically. They can represent various assets, such as:

1. Cryptocurrencies like Bitcoin, Ethereum, and Ripple.
2. Digital representations of physical assets (e.g., real estate, commodities).
3. Utility tokens that grant access or usage rights to a particular platform or service.
4. Security tokens that represent ownership in an asset, such as company shares.

2. Blockchain Technology: Many digital token-based payment systems are built on blockchain technology, a decentralized and immutable ledger. Blockchain ensures transparency, security, and trust in transactions by recording every transaction in a tamper-resistant manner across a distributed network of nodes.

3. Smart Contracts: Smart contracts are self-executing agreements with predefined rules encoded on the blockchain. They automatically execute when certain conditions are met, enabling trustless and automated transactions.

Use Cases:

- 1. Cryptocurrency Payments:** Digital tokens like Bitcoin and Ethereum can be used for everyday transactions, online shopping, and investment.
- 2. Tokenized Assets:** Real-world assets, such as real estate or art, can be tokenized and traded on blockchain platforms, allowing for fractional ownership and increased liquidity.
- 3. Tokenized Securities:** Security tokens represent ownership in a company, fund, or asset. They can streamline the process of buying and selling securities, making it more efficient and accessible.

- 4. Tokenized Loyalty Programs:** Businesses can create loyalty tokens that customers can earn and redeem within a loyalty ecosystem.
- 5. Cross-Border Payments:** Digital tokens enable faster and cheaper cross-border transactions compared to traditional banking systems.
- 6. Micropayments:** Digital tokens are well-suited for micropayments, allowing users to pay small amounts for content or services.

- 4. Wallets:** Users need digital wallets to store and manage their digital tokens. These wallets can be hardware-based, software-based, or even mobile apps, providing secure access to tokens.
- 5. Security:** Security is a critical aspect of digital token-based payment systems. Features like encryption, multi-factor authentication, and private key management are essential to protect users' assets.
- 6. Regulation:** The regulatory environment for digital tokens varies by jurisdiction. Some tokens may be considered securities, and their issuance and trading may be subject to specific regulations.
- 7. Scalability:** Scalability challenges, especially for blockchain-based systems, are being addressed to handle a larger volume of transactions quickly and cost-effectively.
- 8. Interoperability:** To facilitate widespread adoption, some digital token-based systems are working on interoperability solutions to enable tokens from different blockchains to be used interchangeably.

9. Decentralization: Many systems aim for decentralization to reduce reliance on centralized intermediaries, enhance security, and increase transparency. Digital token-based electronic payment systems are part of the broader evolution of finance and commerce. They offer new opportunities for innovation, financial inclusion, and efficiency while also presenting challenges related to regulation, security, and scalability. These systems have gained significant attention and are likely to continue evolving and impacting various industries.

E-Commerce Payment Systems

E-commerce sites use electronic payment, where electronic payment refers to paperless monetary transactions. Electronic payment has revolutionized the business processing by reducing the paperwork, transaction costs, and labor cost. Being user friendly and less time-consuming than manual processing, it helps business organization to expand its market reach/expansion. Listed below are some of the modes of electronic payments –

- Credit Card
- Debit Card
- Smart Card
- E-Money
- Electronic Fund Transfer (EFT)

Credit Card

Payment using credit card is one of most common mode of electronic payment. Credit card is small plastic card with a unique number attached with an account. It has also a magnetic strip embedded in it which is used to read credit card via card readers. When a customer purchases a product via credit card, credit card issuer bank pays on behalf of the customer and customer has a certain time period after which he/she can pay the credit card bill. It is usually credit card monthly payment cycle. Following are the actors in the credit card system.

- The card holder – Customer
- The merchant – seller of product who can accept credit card payments.
- The card issuer bank – card holder's bank
- The acquirer bank – the merchant's bank
- The card brand – for example, visa or Mastercard.

Credit card payment and E- cash are very suitable for _____ e- commerce because the payer and the payee may not have a preestablished relationship, and the payment amount is relatively small.

- (a) B2B
- (b) C2C
- (c) C2B
- (d) B2C

Debit Card

Debit card, like credit card, is a small plastic card with a unique number mapped with the bank account number. It is required to have a bank account before getting a debit card from the bank. The major difference between a debit card and a credit card is that in case of payment through debit card, the amount gets deducted from the card's bank account immediately and there should be sufficient balance in the bank account for the transaction to get completed; whereas in case of a credit card transaction, there is no such compulsion.

Debit cards free the customer to carry cash and cheques. Even merchants accept a debit card readily. Having a restriction on the amount that can be withdrawn in a day using a debit card helps the customer to keep a check on his/her spending.

Smart Card

Smart card is again similar to a credit card or a debit card in appearance, but it has a small microprocessor chip embedded in it. It has the capacity to store a customer's work-related and/or personal information. Smart cards are also used to store money and the amount gets deducted after every transaction.

Smart cards can only be accessed using a PIN that every customer is assigned with. Smart cards are secure, as they store information in encrypted format and are less expensive/provides faster processing. Mondex and Visa Cash cards are examples of smart cards.

E-Money

E-Money transactions refer to situation where payment is done over the network and the amount gets transferred from one financial body to another financial body without any involvement of a middleman. E-money transactions are faster, convenient, and saves a lot of time.

Online payments done via credit cards, debit cards, or smart cards are examples of emoney transactions. Another popular example is e-cash. In case of e-cash, both customer and merchant have to sign up with the bank or company issuing e-cash.

Electronic Fund Transfer

It is a very popular electronic payment method to transfer money from one bank account to another bank account. Accounts can be in the same bank or different banks. Fund transfer can be done using ATM (Automated Teller Machine) or using a computer.

Nowadays, internet-based EFT is getting popular. In this case, a customer uses the website provided by the bank, logs in to the bank's website and registers another bank account. He/she then places a request to transfer certain amount to that account. Customer's bank transfers the amount to other account if it is in the same bank, otherwise the transfer request is forwarded to an ACH (Automated Clearing House) to transfer the amount to other account and the amount is deducted from the customer's account. Once the amount is transferred to other account, the customer is notified of the fund transfer by the bank.

Physical -Needs Disaster and Controls

Physical security is a fundamental aspect of any organization's comprehensive security strategy, protecting personnel, hardware, facilities, and data from a variety of physical threats. It is built on a layered approach designed to deter, detect, delay, and respond to incidents. The specific needs and controls required will vary based on a facility's size, industry, location, and the sensitivity of its assets.

Physical security needs and risks A thorough risk assessment is the foundation for any physical security plan. This process identifies critical assets and the threats against them to determine the necessary controls.

Threats to be addressed:

- **Unauthorized access:** Intruders attempting to gain entry through doors, windows, gates, or by "tailgating" (following an authorized person in).
- **Theft and vandalism:** Malicious acts by external individuals or internal actors (e.g., disgruntled employees) to steal equipment, data, or cause damage.
- **Environmental hazards:** Natural disasters like fires, floods, and earthquakes, as well as man-made accidents like chemical spills or power outages.
- **Sabotage:** Intentional acts to disrupt operations by damaging infrastructure.
- **Human error:** Accidental security breaches caused by employee negligence, such as leaving doors unlocked.

Assets to be protected:

- **Personnel:** Employees, contractors, and visitors.
- **Facilities:** Buildings and grounds.
- **Hardware:** Servers, network devices, and user workstations.
- **Data and information:** Stored on-premise and accessed via network systems.
- **Critical infrastructure:** Essential utilities such as power and telecommunications cabling.

Physical security controls

Physical security controls

Security controls are countermeasures deployed to reduce risk. They are often categorized by their function: preventative, detective, and corrective.

1. Preventative controls

These measures are designed to stop security incidents before they happen.

- **Perimeter security:** Creating a clear boundary for the facility using fencing, gates, walls, and vehicle barriers. This includes adequate security lighting to eliminate hiding spots.
- **Access control:** Restricting and monitoring entry to buildings and sensitive areas. Technologies include key cards, biometric scanners (fingerprint, facial recognition), PIN codes, and secure door locks.

- **Physical barriers:** Using reinforced doors and windows, secure server racks, and locked storage to protect valuable assets.
- **Environmental controls:** Installing fire suppression systems (e.g., sprinklers), flood barriers, and uninterruptible power supplies (UPS) to protect equipment and data from environmental damage.

2. Detective controls

These systems are used to detect and record security events as they occur.

- **Video surveillance (CCTV):** Deploying cameras strategically to monitor high-traffic areas, entry points, and sensitive locations. Modern systems often use Artificial Intelligence (AI) to analyze footage for anomalies, such as tailgating or suspicious behavior.
- **Intrusion detection systems:** Implementing motion sensors, glass-break detectors, and infrared beams that trigger an alarm and notify security personnel when an unauthorized intrusion is detected.
- **Alarms:** Using audible and silent alarms to signal an active breach to both occupants and security staff.
- **Security guards:** Hiring trained personnel for patrolling, monitoring access points, and responding to incidents. Visible security guards also serve as a deterrent.

3. Corrective controls and administrative policies

These measures dictate the response after an incident has been detected.

- **Incident response plan:** A documented plan detailing the steps to be taken in the event of a security breach or emergency, including procedures for lockdowns and involving law enforcement.
- **Employee training:** Regular training for employees on physical security protocols, including how to handle visitors, prevent tailgating, and respond to emergencies.
- **Regular audits and testing:** Frequent audits and penetration testing to identify and address vulnerabilities in the security system. Fire drills and emergency simulations are also essential for training employees on response protocols.
- **Maintenance:** Regularly inspecting and updating all physical security hardware and software, from locks to access control firmware, to ensure they function properly and are protected against new threats.
- **Clear accountability:** Establishing clear roles and responsibilities for security, from managing data center access to overseeing facility-wide security measures.

Basic Tenets of Physical Security and Physical Entry Controls, Access Control

Basic Tenets of Physical Security and Physical Entry Controls, Access Control

Introduction to Physical Security

These days, tips on how to strengthen your cyber security follow the announcement of every another cyber attack. Don't forget to backup your data, apply patches over vulnerabilities, monitor firewalls, etc. It is very important to remember that software is not your only weapon when it comes to cyber security. Physical Cyber Security is another tier in your line of defense.

According to Goldstein(2016), Physical Security is critical, especially for small business that does not have many resources to devote to security personnel and tools as opposed to larger firms. When it comes to Physical Security, the same principles apply here:

- Identify and classify your assets and resources.
- Identify plausible threats.
- Identify probable vulnerabilities that threats may exploit.
- Identify the expected cost in case if an attack occurs.

Factors on which Physical Security Depends

1. How many workplaces, buildings or sites are there in an organization?
2. Size of the building of the organization?
3. How many employees are employed in the organization?
4. How many entry and exit points are there in the organization?
5. Points of placement of data centers and other confidential information.

Layers of Physical Security

Layers in Physical Security are implemented at the perimeter and are moving towards an asset. The layers are as follows:

1. Deterrence

The goal of Deterrence methods is to convince a potential attacker that a successful attack is not possible due to strong defenses. For example: By placing your keys inside a highly secure key control system made up of heavy metal like steel, you can help prevent attackers from gaining access to assets. *Deterrence methods are classified into 4 categories:*

- **Physical Barriers:** These include fences, walls, vehicle barriers, etc. They also act as a Psychological deterrent by defining the perimeter of the facility and making intrusion seem more difficult.
- **Combination Barriers:** These are designed to defeat defined threats. This is a part of building codes as well as fire codes.

- **Natural Surveillance:** In this architects seek to build places that are more open and visible to authorized users and security personnel so that attackers are unable to perform the unauthorized activity without being seen. For example-decreasing the amount of dense and tall vegetation.
- **Security Lighting:** Doors, gates or other means of the entrance should be well lit as Intruders are less likely to enter well-lit areas. Keep mind to place lighting in a manner, that is difficult to tamper.

2. Detection : If you are using the manual key control system, you have no way of knowing the exact timestamp of when an unauthorized user requested a key or has exceeded its time limit. Detection methods can be of the following types:

- **Alarm Systems and Sensors:** Alarm systems can be installed to alert security personnel in case of an attempt of unauthorized access. They consist of sensors like perimeter sensors, motion sensors, etc.
- **Video Surveillance:** Surveillance cameras can be used for detection if an attack has already occurred and a camera is placed at the point of attack. Recorded video can be used

3. Access Control

These methods are used to monitor and control the traffic through specific access points. Access Control includes the following methods:

- **Mechanical Access Control Systems:** These includes gates, doors, locks, etc.
- **Electronic Access Control:** These are used to monitor and control larger populations, controlling for user life cycles, dates and individual access points.
- **Identification System and access policies:** These includes the use of policies, procedures and processes to manage the access into the restricted area.

4. Security Personnel

They play a central role in all layers of security. They perform many functions like:

- Administering electronic access control.
- Responding to alarms.
- Monitoring and analyzing video footage and many more

Countermeasures and Protection Techniques

1. Protection against Dumpster Diving

Dumpster Diving is the process of finding some useful information about the person or business from the trash that can later be used for hacking purpose. Since the information is in the trash, it is not useful for the owner but deemed useful to the picker. To protect against it, you need to follow certain measures:

- Ensure all important documents are shredded and they are still secure.
- Destroy any CDs/ DVDs containing personal data.
- Make sure that nobody can walk into your building and simply steal your garbage and should have safe disposal policy.
- Firewalls can be used to prevent suspicious users from accessing the discarded data.

2. Employee Awareness Training

A negligent employee can be one of the major causes of a Cyber security breach. Employee awareness training sessions can help in such cases. Employee awareness training should focus on one underlying theme- *avoid the SEP- Somebody else's problem field.*

3. Site Access Control

Lack of Access Control can be highly devastating if a wrong person gets in and gets access to sensitive information. Fortunately nowadays, you have a number of modern tools that will help you to optimize your access control.

- **Envoy** is a tool that will help you to expand access to guests in controlled manner.
- **Open Path** is a mobile system that allows access to only a limited set of people within the directory using smartphones and other devices.

4. Securing Your Windows

If you have the data that hackers would love to get their hands on, they will try any method and might just look through the window. Make sure you are aware of the sight angles to position your screens and other devices. Overlooking from different sight angles to see your credentials is known as *Shoulder Surfing*.

5. Secure Network-Enabled Printers

Network Printers are a very convenient option allowing anyone in the office to get connected, without a need of extra wiring. Unfortunately, they have underlying security risks also. Sometimes, due to default settings, they offer open WiFi access, thus allowing anyone to get in and open vulnerabilities in the process.

- Only connect those to the Internet that actually needs to be.
- Remote access is not necessary for scenarios where only people from your office use the printer.
- You can add passwords to the connection if necessary.

6. Securing Your Backups

Physical backups are critical for business continuity, helping you prevent data loss in the event of disasters, outages, and more. Most businesses secure their servers but they forget that backups are equally important. They are holding the same level of sensitive data as servers. Treat your backups as you treat your sensitive information and secure them.

7. Building Secure Guest Wifi

Guest WiFi is a natural solution when you have guests or visitors. Here are a few tricks to help protect your resources from the external users:

- Segment your network- In this way, it isolates Guest WiFi from your internal devices and data.
- Encrypt your wireless signals and change the default passwords of all devices on the network.

8. Locking up your Servers

Any area in your organization that stores data need to be secured. Locking doors and making sure server area gets extra protection.

9. Accounting for Loss or Stolen Devices

As devices are becoming more mobile, chances for them being stolen or falling out of someone's pocket becomes more frequent. Mobile Device Management can help you to manage such situations and take the necessary precautions. The best solution in such cases is to simply lock down and potentially wipe any lost or stolen devices from the organization remotely.

10. Implementing video systems

To achieve a more secure premises, it is advisable to use a Video Surveillance system.

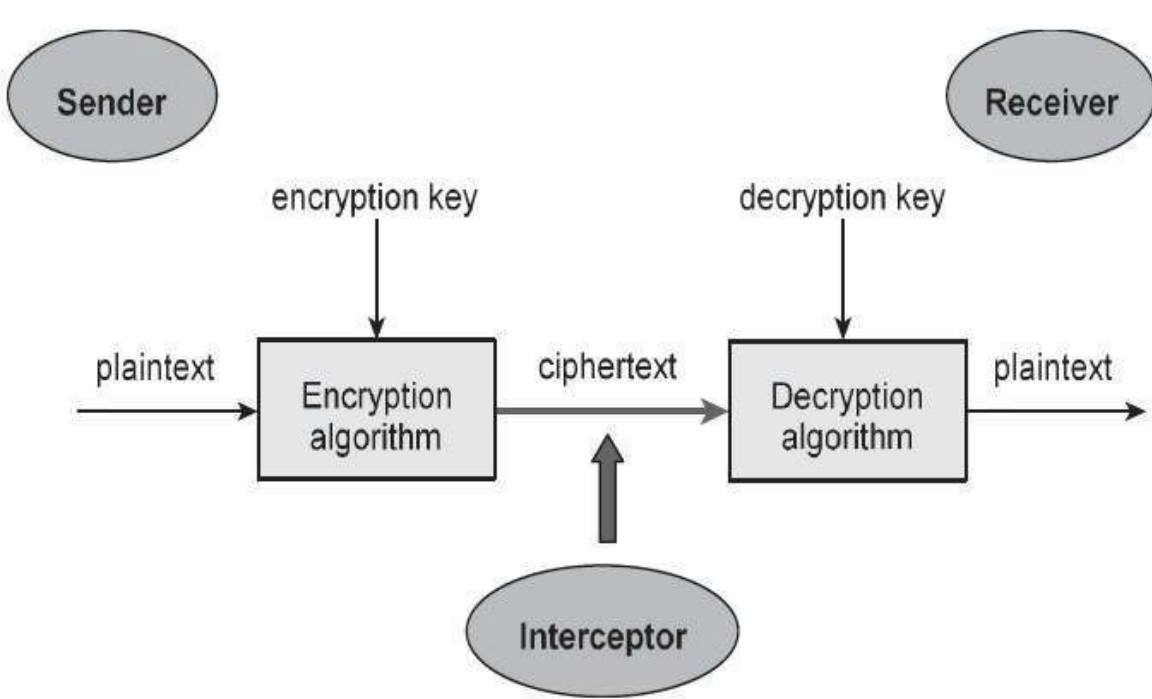
- Mere presence of cameras can deter potential attackers.
- Availability of video footage allows you to have continuous monitoring over the entire premises.
- If an attack happens, you can check the recorded video, easily reconcile the process and catch the perpetrator.

Model of Cryptographic Systems, Design and Implementation Issues, Policies

CRYPTOSYSTEMS

A cryptosystem is an implementation of cryptographic techniques and their accompanying infrastructure to provide information security services. A cryptosystem is also referred to as a cipher system.

Let us discuss a simple model of a cryptosystem that provides confidentiality to the information being transmitted. This basic model is depicted in the illustration below –



The illustration shows a sender who wants to transfer some sensitive data to a receiver in such a way that any party intercepting or eavesdropping on the communication channel cannot extract the data.

The objective of this simple cryptosystem is that at the end of the process, only the sender and the receiver will know the plaintext.

Components of a Cryptosystem

The various components of a basic cryptosystem are as follows –

- **Plaintext** : It is the data to be protected during transmission. Encryption Algorithm. It is a mathematical process that produces a ciphertext for any given plaintext and encryption key. It is a cryptographic algorithm that takes plaintext and an encryption key as input and produces a ciphertext.
- **Ciphertext** : It is the scrambled version of the plaintext produced by the encryption algorithm using a specific the encryption key. The ciphertext is not guarded. It flows on public channel. It can be intercepted or compromised by anyone who has access to the communication channel.
- **Decryption Algorithm** : It is a mathematical process, that produces a unique plaintext for any given ciphertext and decryption key. It is a cryptographic algorithm that takes a ciphertext and a decryption key as input, and outputs a plaintext. The decryption algorithm essentially reverses the encryption algorithm and is thus closely related to it.

- **Encryption Key** : It is a value that is known to the sender. The sender inputs the encryption key into the encryption algorithm along with the plaintext in order to compute the ciphertext.
- **Decryption Key** : It is a value that is known to the receiver. The decryption key is related to the encryption key, but is not always identical to it. The receiver inputs the decryption key into the decryption algorithm along with the ciphertext in order to compute the plaintext.

For a given cryptosystem, a collection of all possible decryption keys is called a key space. An interceptor or an attacker is an unauthorized entity who attempts to determine the plaintext. He can see the ciphertext and may know the decryption algorithm. He, however, must never know the decryption key.

Types of Cryptosystems

Fundamentally, there are two types of cryptosystems based on the manner in which encryption decryption is carried out in the system –

- Symmetric Key Encryption
- Asymmetric Key Encryption

The main difference between these cryptosystems is the relationship between the encryption and the decryption key. Logically, in any cryptosystem, both the keys are closely associated. It is practically impossible to decrypt the ciphertext with the key that is unrelated to the encryption key.

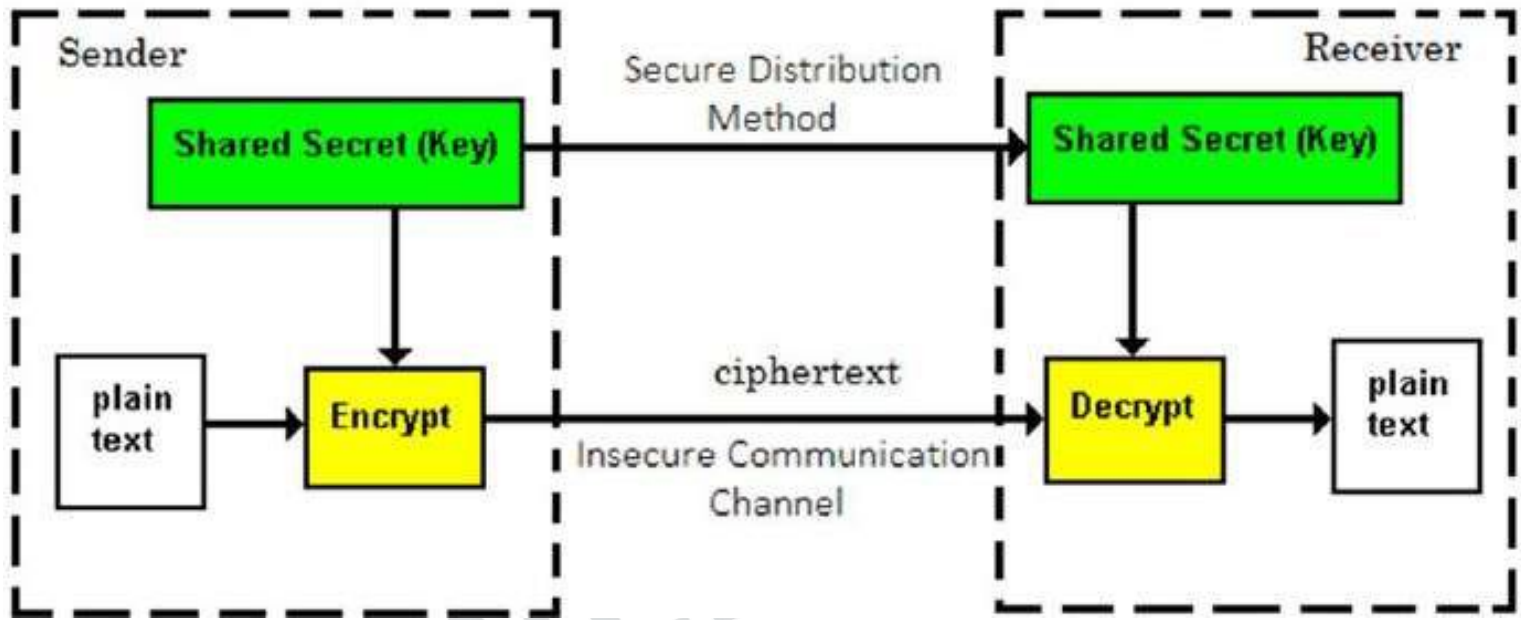
Symmetric Key Encryption

The encryption process where same keys are used for encrypting and decrypting the information is known as Symmetric Key Encryption.

The study of symmetric cryptosystems is referred to as symmetric cryptography.

Symmetric cryptosystems are also sometimes referred to as secret key cryptosystems.

A few well-known examples of symmetric key encryption methods are – Digital Encryption Standard DES, Triple-DES 3DES, IDEA, and BLOWFISH.



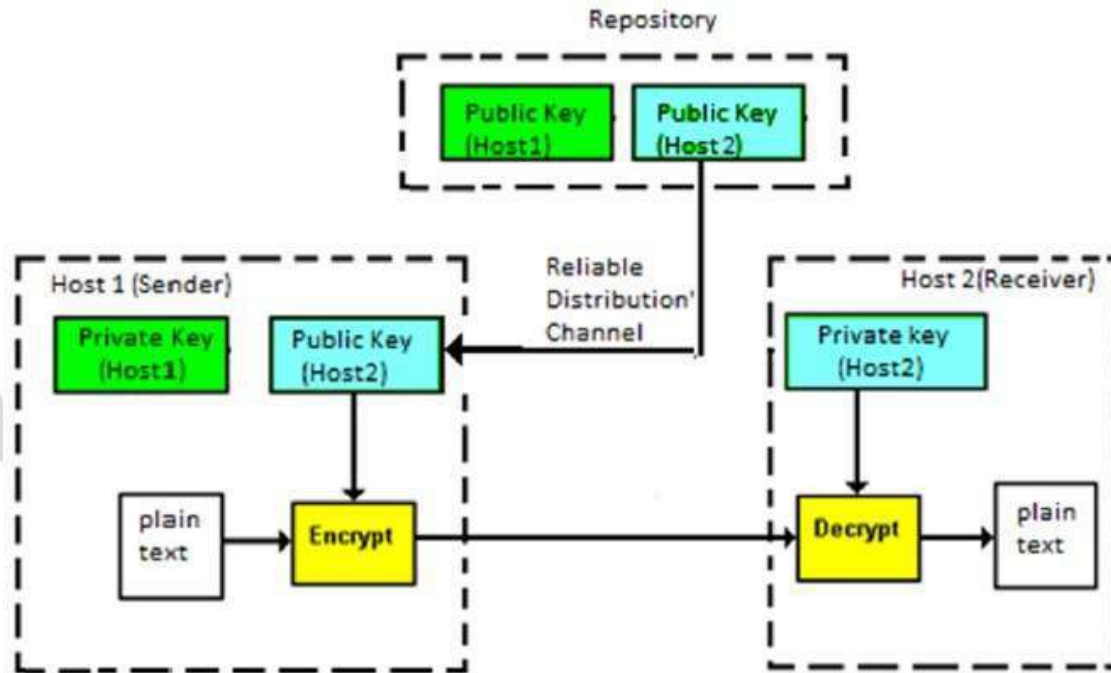
Prior to 1970, all cryptosystems employed symmetric key encryption. Even today, its relevance is very high and it is being used extensively in many cryptosystems. It is very unlikely that this encryption will fade away, as it has certain advantages over asymmetric key encryption.

The salient features of cryptosystem based on symmetric key encryption are –

- Persons using symmetric key encryption must share a common key prior to exchange of information.
- Keys are recommended to be changed regularly to prevent any attack on the system.
- A robust mechanism needs to exist to exchange the key between the communicating parties. As keys are required to be changed regularly, this mechanism becomes expensive and cumbersome.
- In a group of n people, to enable two-party communication between any two persons, the number of keys required for group is $n \times (n-1)/2$.
- Length of Key number of bits in this encryption is smaller and hence, process of encryption decryption is faster than asymmetric key encryption.
- Processing power of computer system required to run symmetric algorithm is less.

Asymmetric Key Encryption

The encryption process where different keys are used for encrypting and decrypting the information is known as Asymmetric Key Encryption. Though the keys are different, they are mathematically related and hence, retrieving the plaintext by decrypting ciphertext is feasible. The process is depicted in the following illustration –



Asymmetric Key Encryption was invented in the 20th century to come over the necessity of pre shared secret key between communicating persons. The salient features of this encryption scheme are as follows –

- Every user in this system needs to have a pair of dissimilar keys, private key and public key. These keys are mathematically related – when one key is used for encryption, the other can decrypt the ciphertext back to the original plaintext.
- It requires to put the public key in public repository and the private key as a well-guarded secret. Hence, this scheme of encryption is also called Public Key Encryption.
- Though public and private keys of the user are related, it is computationally not feasible to find one from another. This is a strength of this scheme.
- When Host1 needs to send data to Host2, he obtains the public key of Host2 from repository, encrypts the data, and transmits.
- Host2 uses his private key to extract the plaintext.
- Length of Keys number of bits in this encryption is large and hence, the process of encryption decryption is slower than symmetric key encryption.
- Processing power of computer system required to run asymmetric algorithm is

Symmetric cryptosystems are a natural concept. In contrast, public-key cryptosystems are quite difficult to comprehend.

You may think, how can the encryption key and the decryption key are 'related', and yet it is impossible to determine the decryption key from the encryption key? The answer lies in the mathematical concepts. It is possible to design a cryptosystem whose keys have this property. The concept of public-key cryptography is relatively new. There are fewer public-key algorithms known than symmetric algorithms.

Symmetric Ciphers

Symmetric ciphers use a single shared secret key for both encryption and decryption.

- Caesar Cipher (Shift Cipher): A simple, historical substitution cipher.
- Playfair Cipher: An early manual block cipher operating on digraphs.
- Data Encryption Standard (DES): A former US national standard block cipher, now considered insecure for most uses due to its small 56-bit key size.
- Triple DES (3DES): A more secure variation of DES that applies the algorithm three times, often found in financial systems.
- Advanced Encryption Standard (AES): The current widely used standard block cipher with key lengths of 128, 192, or 256 bits, based on the Rijndael algorithm.
- Blowfish: A block cipher known for its speed and efficiency, created by Bruce Schneier.
- Twofish: Another block cipher that was an AES finalist, offering strong security and flexibility.
- International Data Encryption Algorithm (IDEA): A block cipher using a 128-bit key.
- RC4: A widely used stream cipher.

Asymmetric Ciphers (Public-Key Cryptography)

Asymmetric ciphers use a pair of keys: a public key for encryption and a private key for decryption or digital signing.

- **RSA (Rivest-Shamir-Adleman):** One of the first and most widely used public-key cryptosystems, used for both encryption and digital signatures, based on the difficulty of factoring large integers.
- **ECC (Elliptic Curve Cryptography):** Provides strong security with smaller key sizes compared to RSA, based on the algebraic structure of elliptic curves.
- **DSA (Digital Signature Algorithm):** Used for digital signatures, often mentioned in conjunction with the Digital Signature Standard (DSS).
- **Diffie-Hellman (DH):** A key exchange algorithm that allows two parties to securely establish a shared secret key over an insecure communication channel.
- **ElGamal:** An asymmetric encryption system based on discrete logarithms.

Which of the following is NOT an example of symmetric key encryption?

- (a) Digital encryption standard (DES)
- (b) Triple-DES (3DES)
- (c) RSA
- (d) BLOWFISH

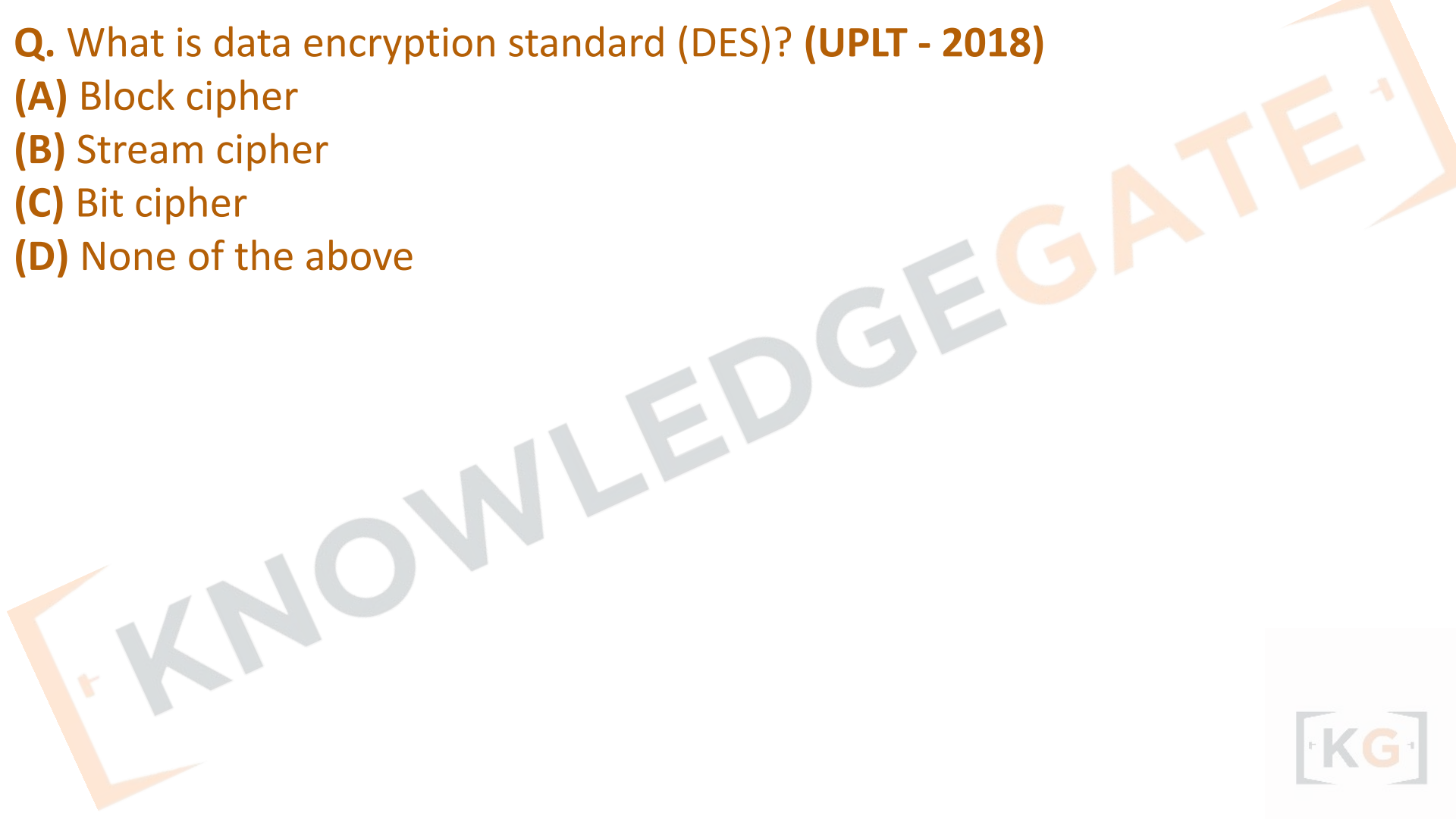
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Ans. ©



Q. What is data encryption standard (DES)? (UPLT - 2018)

- (A) Block cipher**
- (B) Stream cipher**
- (C) Bit cipher**
- (D) None of the above**



Public key encryption makes use of

- (a) one key
- (b) two keys
- (c) hash function
- (d) All the given options

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Ans. (b)



What is the total number of required keys if N individuals need to communicate using public key cryptography?

- (a) $N-1$
- (b) $N+1$
- (c) $2N$
- (d) $N(N-1)/2$

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Ans. (c)



In which of the following cases, does the attacker try to determine the key by attempting all possible keys?

- (a) Dictionary attack
- (b) Man in middle attack
- (c) Brute force attack
- (d) Timing attack

Intrusion Detection System (IDS)

Intrusion is when an attacker gets unauthorized access to a device, network, or system. Cyber criminals use advanced techniques to sneak into organizations without being detected.

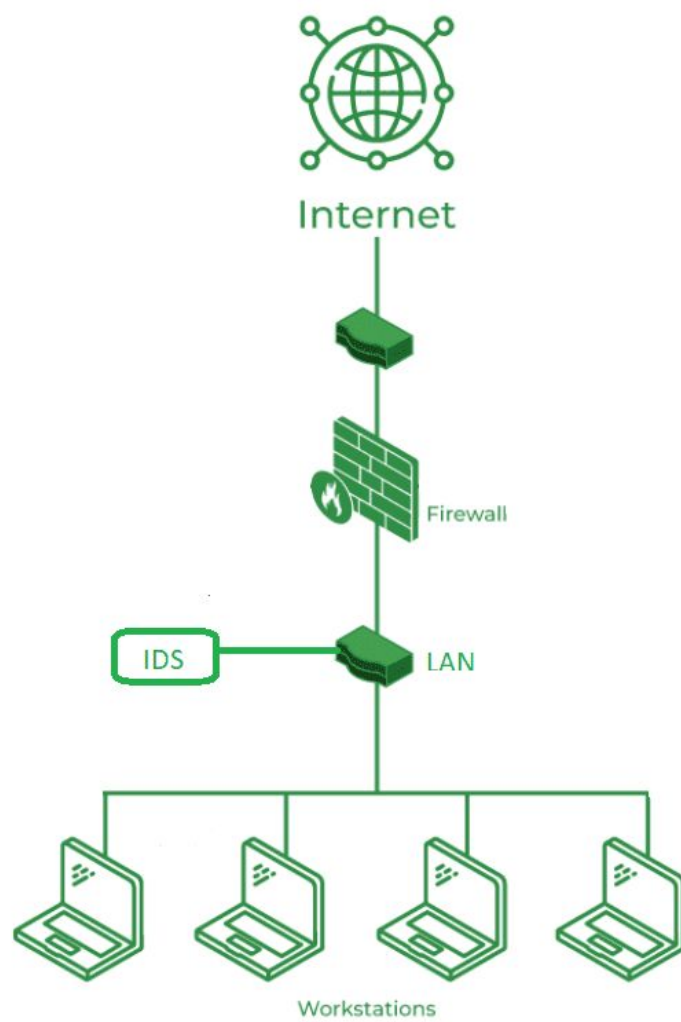
Intrusion Detection System (IDS) observes network traffic for malicious transactions and sends immediate alerts when it is observed. It is software that checks a network or system for malicious activities or policy violations. Each illegal activity or violation is often recorded either centrally using an SIEM system or notified to an administration. IDS monitors a network or system for malicious activity and protects a computer network from unauthorized access from users, including perhaps insiders. The intrusion detector learning task is to build a predictive model (i.e. a classifier) capable of distinguishing between 'bad connections' (intrusion/attacks) and 'good (normal) connections'.

Common Methods of Intrusion

- **Address Spoofing:** Hiding the source of an attack by using fake or unsecured proxy servers making it hard to identify the attacker.
- **Fragmentation:** Sending data in small pieces to slip past detection systems.
- **Pattern Evasion:** Changing attack methods to avoid detection by IDS systems that look for specific patterns.
- **Coordinated Attack:** Using multiple attackers or ports to scan a network, confusing the IDS and making it hard to see what is happening.

Working of Intrusion Detection System(IDS)

- An IDS (Intrusion Detection System) monitors the traffic on a computer network to detect any suspicious activity.
- It analyzes the data flowing through the network to look for patterns and signs of abnormal behavior.
- The IDS compares the network activity to a set of predefined rules and patterns to identify any activity that might indicate an attack or intrusion.
- If the IDS detects something that matches one of these rules or patterns, it sends an alert to the system administrator.
- The system administrator can then investigate the alert and take action to prevent any damage or further intrusion.



Classification of Intrusion Detection System(IDS)

Intrusion Detection System are classified into 5 types:

- **Network Intrusion Detection System (NIDS):** Network intrusion detection systems (NIDS) are set up at a planned point within the network to examine traffic from all devices on the network. It performs an observation of passing traffic on the entire subnet and matches the traffic that is passed on the subnets to the collection of known attacks. Once an attack is identified or abnormal behavior is observed, the alert can be sent to the administrator. An example of a NIDS is installing it on the subnet where firewalls are located in order to see if someone is trying to crack the firewall.

- **Host Intrusion Detection System (HIDS):** Host intrusion detection systems (HIDS) run on independent hosts or devices on the network. A HIDS monitors the incoming and outgoing packets from the device only and will alert the administrator if suspicious or malicious activity is detected. It takes a snapshot of existing system files and compares it with the previous snapshot. If the analytical system files were edited or deleted, an alert is sent to the administrator to investigate. An example of HIDS usage can be seen on mission-critical machines, which are not expected to change their layout.

- **Hybrid Intrusion Detection System:** Hybrid intrusion detection system is made by the combination of two or more approaches to the intrusion detection system. In the hybrid intrusion detection system, the host agent or system data is combined with network information to develop a complete view of the network system. The hybrid intrusion detection system is more effective in comparison to the other intrusion detection system. Prelude is an example of Hybrid IDS.

- **Application Protocol-Based Intrusion Detection System (APIDS):** An application Protocol-based Intrusion Detection System (APIDS) is a system or agent that generally resides within a group of servers. It identifies the intrusions by monitoring and interpreting the communication on application-specific protocols. For example, this would monitor the SQL protocol explicitly to the middleware as it transacts with the database in the web server.

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- **Protocol-Based Intrusion Detection System (PIDS):** It comprises a system or agent that would consistently reside at the front end of a server, controlling and interpreting the protocol between a user/device and the server. It is trying to secure the web server by regularly monitoring the HTTPS protocol stream and accepting the related HTTP protocol. As HTTPS is unencrypted and before instantly entering its web presentation layer then this system would need to reside in this interface, between to use the HTTPS.
- **Signature-Based Detection:** Signature-based detection checks network packets for known patterns linked to specific threats. A signature-based IDS compares packets to a database of attack signatures and raises an alert if a match is found. Regular updates are needed to detect new threats, but unknown attacks without signatures can bypass this system

Intrusion Detection System Evasion Techniques

- **Fragmentation:** Dividing the packet into smaller packet called fragment and the process is known as fragmentation. This makes it impossible to identify an intrusion because there can't be a malware signature.
- **Packet Encoding:** Encoding packets using methods like Base64 or hexadecimal can hide malicious content from signature-based IDS.
- **Traffic Obfuscation:** By making message more complicated to interpret, obfuscation can be utilised to hide an attack and avoid detection.
- **Encryption:** Several security features such as data integrity, confidentiality, and data privacy, are provided by encryption. Unfortunately, security features are used by malware developers to hide attacks and avoid detection.

Detection Method of IDS

- **Signature-Based Method:** Signature-based IDS detects the attacks on the basis of the specific patterns such as the number of bytes or a number of 1s or the number of 0s in the network traffic. It also detects on the basis of the already known malicious instruction sequence that is used by the malware. The detected patterns in the IDS are known as signatures. Signature-based IDS can easily detect the attacks whose pattern (signature) already exists in the system but it is quite difficult to detect new malware attacks as their pattern (signature) is not known.

- **Anomaly-Based Method:** Anomaly-based IDS was introduced to detect unknown malware attacks as new malware is developed rapidly. In anomaly-based IDS there is the use of machine learning to create a trustful activity model and anything coming is compared with that model and it is declared suspicious if it is not found in the model. The machine learning-based method has a better-generalized property in comparison to signature-based IDS as these models can be trained according to the applications and hardware configurations.

Comparison of IDS with Firewalls

IDS and firewall both are related to network security but an IDS differs from a firewall as a firewall looks outwardly for intrusions in order to stop them from happening. Firewalls restrict access between networks to prevent intrusion and if an attack is from inside the network it doesn't signal. An IDS describes a suspected intrusion once it has happened and then signals an alarm.

Why Are Intrusion Detection Systems (IDS) Important?

An Intrusion Detection System (IDS) adds extra protection to your cybersecurity setup, making it very important. It works with your other security tools to catch threats that get past your main defenses. So, if your main system misses something, the IDS will alert you to the threat.

Benefits of IDS

- **Detects Malicious Activity:** IDS can detect any suspicious activities and alert the system administrator before any significant damage is done.
- **Improves Network Performance:** IDS can identify any performance issues on the network, which can be addressed to improve network performance.
- **Compliance Requirements:** IDS can help in meeting compliance requirements by monitoring network activity and generating reports.
- **Provides Insights:** IDS generates valuable insights into network traffic, which can be used to identify any weaknesses and improve network security.

Disadvantages of IDS

- **False Alarms:** IDS can generate false positives, alerting on harmless activities and causing unnecessary concern.
- **Resource Intensive:** It can use a lot of system resources, potentially slowing down network performance.
- **Requires Maintenance:** Regular updates and tuning are needed to keep the IDS effective, which can be time-consuming.
- **Doesn't Prevent Attacks:** IDS detects and alerts but doesn't stop attacks, so additional measures are still needed.
- **Complex to Manage:** Setting up and managing an IDS can be complex and may require specialized knowledge.

Security metrics- Classification and their benefits

Cyber Security Metrics

Metrics are tools to facilitate decision-making and improve performance and accountability. A cybersecurity metric contains the number of reported incidents, any fluctuations in these numbers as well as the identification time and cost of an attack. Thus, it provides stats that can be used to ensure the security of the current application.

Organizations get the overall view of threats in terms of time, severity, and number. It is important today when this data keeps fluctuating. This way the organizations can maximize protection from threats in the future.

Use of a Cybersecurity Metric:

A Cybersecurity metric assists the organization in the following ways:

- It facilitates decision-making and improves overall performance and accountability.
- It helps in setting quantifiable measures based on objective data in the metric.
- It helps in making corrections in an efficient way.
- It brings together all the factors like finance, regulation, and organization to measure security.
- It maintains the log of every individual system that has been tested over the years.

Some Cybersecurity Metrics:

Here is a list of some important cybersecurity metrics that portray the current threat scenario really well.

- **A number of systems have vulnerabilities:** A very important cybersecurity metric is to know where your assets lag. This helps in determining risks along with the improvements that must be taken. This way the vulnerabilities can be worked upon before anyone exploits them.

Some Cybersecurity Metrics:

- **Mean detection and response time:** The sooner a cybersecurity breach is detected and responded to, the lesser will be the loss. It is important to have systems that reduce the mean detection and response time.
- **Data volume over a corporate network:** Employees having unrestricted access to the company's internet may turn out into a disaster. If they use the company's resources to download anything, it might lead to the invasion of malware.

Some Cybersecurity Metrics:

- **Incorrectly configured SSL certificates:** Company's digital identity can be used to extract critical information if proper authentication measures are not in place. Thus, it is important to keep track of SSL certificates that are not correctly configured.
- **Deactivation time of credentials of a former employee:** The employees no longer a part of the organization must not be given access to the company's resources. Moreover, their previous rights must be immediately terminated otherwise sensitive information might be put at risk

Some Cybersecurity Metrics:

- **The number of users having higher access levels:** There are individuals that have a wider range of data access as compared to others. However, this all must be efficiently monitored by the company. Also, unnecessary access should be minimized.
- **Open communication ports during a time period:** Communication occurs both ways. The ports for inbound and outbound traffic must be individually monitored. NetBIOS must be avoided in inbound traffic and SSL should be rightly monitored in outbound traffic. Also, ports that allow protocols for remote sessions must be monitored for a period of time.

Some Cybersecurity Metrics:

- **Access to systems by third parties:** Some systems of a company are more critical to others. For the critical ones, proper mapping of third parties using them should be monitored.
- **Review of frequency of third party access:** Third parties might have to access the network of a company to complete any project or activity. Thus, monitoring their access is important to identify any suspicious activity that might be undergoing at their end.

Advantage of using metrics.

- **For learning:** To figure out different information pertaining to a system, we have to start by asking questions. These questions will lead us to answers and then in turn to information. This becomes easier with the help of a metric and thus the understanding of cybersecurity risks improves.
- **For Decision Making:** When we use a metric to gain information about a system, we can extend its use even further by gaining insight into previous decisions. This way, we can better manage the decisions that have to be taken with respect to current cybersecurity risks.
- **For Implementation of Plans:** After analyzing the loopholes in the system and making decisions on how to go about rectifying them, it is time to take action. This implementation can be supported further by referring to previous records and assessments in the cybersecurity metric.

Cyber Safety and Security

Introduction : Nowadays technology is growing very fast; we use the internet for searching information, entertainment and talking to people. When we use digital media or internet, then we leave digital footprints. However, it's crucial to understand potential risks like cybercrimes, trolling, cyberbullying, etc., and be equipped with the knowledge and tools to stay safe.

Cyber safety and cybersecurity are both about staying safe online, but they focus on different things:

- **Cyber safety:** Cyber safety is about protecting yourself from online threats like malware, phishing attacks, cyberbullying and inappropriate content.
- **Cyber security:** Cyber security is about protecting information systems and data from unauthorised access, use, disclosure, disruption, modification, or destruction. This focuses on protecting systems and networks from attacks.

Digital Footprints

What are Digital Footprints?

Digital footprint is also known as digital shadow or electronic footprint. For example, if you walk in sand, then you leave footprints in the sand. In the same manner, when you use the internet, then each and every click on internet, every search, and every message you send create your online record, which is known as a digital footprint.

Digital footprints can be classified into two categories.

Active footprint

Passive footprint

Active footprint

An active data footprint refers to the data that is deliberately provided by an individual through their online activities.

This includes:

- When you share information like your name, email address, phone number and other personal details on social media, forms, surveys, etc.
- When you are searching queries, sending email to others, or sharing, uploading or storing photos or videos.
- You can give comments on posts, create online accounts or make online purchases.

Passive footprint

A passive data footprint refers to the data that is collected automatically through an individual's online activities, without their intentional participation.

This includes:

- Data such as browsing history, device information, IP address, location
- Websites you visit (tracked through cookies)
- Date and time a photo was taken, the location where it was taken, and the device that was used to take it
- Data that is collected by websites, apps, and other online services.

Data Privacy

Data privacy means keeping your personal information safe and deciding who can see it and how it is used. For example, suppose you have a secret diary and you want to share the information with others. If you want to protect your information, then you have to understand who can read your diary and what to share and keep it safe from strangers. That's exactly what data privacy does.

Definition: Data privacy ensures that personal information is not misused or abused by third parties. This includes protecting the data from unauthorised access, theft, or other types of misuse.

Understanding Data Privacy in the Digital Age

In today's world, data is money. Every time you click, swipe or search on the internet, you leave a digital footprint. This footprint contains personal information that can be collected, analysed and used by many companies and organisations; it means that your data are being tracked by the organisation, which makes it harder to protect your privacy.

Why is data privacy important?

- **Your online reputation:** Everything you share online builds your virtual image. Schools, employers, and even friends can look at your data to form an impression of you.
- **Targeted advertising:** Companies use your data to personalize the ads you see, which can sometimes feel intrusive.
- **Protecting yourself from harm:** Sharing too much information online can make you vulnerable to scams, cyberbullying, and even identity theft.

How can you protect your data privacy?

- **Be mindful of what you share online:** Think twice before posting anything, especially personal information.
- **Adjust your privacy settings:** Every website and app have privacy settings. Learn how to control who sees your information and how it's used.
- **Use strong passwords and keep them secret:** Don't share your passwords with anyone.
- **Beware of phishing scams:** Don't click on suspicious links or open emails from unknown senders.
- **Be cautious about using public Wi-Fi:** Avoid sensitive activities like online banking on public Wi-Fi networks.

- **Learn about your rights:** You have the right to know what data is collected about you and how it's used.
- **Stronger Regulations:** Governments need to enact and enforce robust data protection laws, setting clear standards for data collection, use, and storage. This includes granting individuals the right to access, rectify, and erase their personal data.
- **Ethical Practices:** Organizations must prioritize ethical data practices by being transparent about data collection, obtaining informed consent, and implementing robust security measures to protect user data.

How to Protect Personal Data and Stay Safe Online

- VPNs (Virtual Private Networks) hide your location and browsing activity.
- Encrypted messaging apps keep your chats private and secure.

Laws and regulations to protect your data

- **Information Technology (Reasonable Security Practices and Procedures and Sensitive Personal Data or Information) Rules, 2011:** This rule protects sensitive personal data like financial details, passwords and health records.
- **The Aadhaar (Targeted Delivery of Financial and Other Subsidies, Benefits and Services) Act, 2016:** which regulates the collection and use of biometric data for the purpose of authentication.
- **Personal Data Protection Bill (PDPB), 2019:** create a comprehensive framework for data privacy and give individuals rights like accessing their data, correcting inaccuracies, and withdrawing consent.

Data Security

What Is Data Security? Data security is like locking your digital doors and windows; it helps to protect your personal information from hackers, cybercriminals and unauthorised access. Whether it is your personal information like name, address, or password, keeping this data safe is essential in today's connected world.

Key Pillars of Data Security

Confidentiality: Ensuring that data is accessible only to authorised individuals or systems, like encryption, access controls and data classification.

Integrity: Integrity helps to stay accurate and unchanged, like checksums, digital signatures and validation techniques.

Authorisation: Appropriate access permissions to users based on their roles, responsibilities, and levels of trust, like role-based permissions and access control lists.

Encryption: Encoding data in such a way that it can only be decrypted and read by authorised parties with the proper cryptographic keys, like cryptographic keys, secure transmission and storage.

Importance of Data Security

- **Protects personal privacy:** Ensures that sensitive information remains confidential and is not misused.
- **Prevents identity theft:** Safeguards against unauthorized access to personal accounts and financial information.
- **Maintains trust:** Helps build trust and confidence in online transactions and communication.
- **Preserves reputation:** Avoids potential damage to reputation or embarrassment from leaked personal information.

Security Measures:

- **Strong Passwords:** Always use complex combinations of letters, numbers and symbols. Never share passwords, even with friends.
- **Privacy Settings:** Adjust privacy settings on social media platforms and online accounts to control who can see personal information.
- **Secure connections:** Ensure that websites use HTTPS encryption for secure communication and avoid connecting to unsecured Wi-Fi networks.
- **Antivirus software :** Install reputable antivirus software to detect and remove malware from computer systems.
- **Two-factor authentication (2FA):** Enable 2FA for added security by requiring a second form of verification, such as a text message or authentication app, in addition to a password.
- **Safe Browsing:** Be cautious when clicking on links or downloading files from unknown or suspicious websites. Use reputable antivirus software to protect against malware and viruses.

- **Email Safety:** Be wary of phishing emails that may try to trick you into revealing personal information or clicking on malicious links.
- **Data Backup:** Regularly back up important files and schoolwork to an external hard drive, USB flash drive, or cloud storage service like Google Drive or Dropbox.
- **Social Media and Online Interaction:** Be mindful of what you post online and how it may affect your reputation and future opportunities.
- **Respect Copyright:** Avoid using copyrighted materials (such as images, music, or text) without permission. Instead, look for creative commons or public domain content that can be freely used and shared.

Responsible Online Behaviour

Think before sharing: Be cautious about sharing personal information online and avoid disclosing sensitive details to strangers.

Verify sources: Verify the credibility of websites and sources before trusting or sharing information found online.

Report suspicious activity: Report any suspicious emails, messages or websites to trusted adults or authorities.

Intellectual Property

Intellectual property rights (IPR) are the legal protections granted to the creators of original works and inventions. Intellectual property (IP) refers to creations of the human mind, including:

- **Inventions:** New products, processes, or devices (protected by patents).
- **Literary and artistic works:** books, music, paintings, software (protected by copyrights).
- **Designs:** Industrial designs, product packaging (protected by design rights).
- **Symbols, names, and images:** Trademarks and logos (protected by trademarks).

Understanding intellectual property is important for various reasons:

- **Creative professionals:** Knowing how to protect their own work and how to use the work of others legally.
- **Businesses:** Protecting their brand identity and innovations while understanding how to use existing intellectual property responsibly.
- **Consumers:** Being aware of the rights associated with intellectual property and respecting them.

Plagiarism

When you are copying any other person's information and giving your name, it is known as plagiarism. Plagiarism is the act of using someone else's words, ideas, or creative work without giving credit to them.

Direct Plagiarism: if any person is copying entire passages, sentences or paragraphs without citation, it is known as direct plagiarism.

Paraphrasing Plagiarism: Paraphrasing plagiarism involves when you rephrase someone else's ideas or words without proper citation.

Mosaic Plagiarism (Patchwriting): When you mix copied phrases or sentences from different sources into your own writing without proper citation.

Accidental Plagiarism: Accidental plagiarism occurs when a person unintentionally fails to properly cite or attribute sources in their work.

Inappropriate Citation: Inappropriate citation involves citing a source incorrectly, misleadingly, or without actually consulting it.

Verbatim Plagiarism: Verbatim plagiarism involves directly copying someone else's work word for word without any changes and without proper citation.

KNOWLEDGE



How to prevent plagiarism?

- Understand proper citation: Acknowledge all sources used.
- Paraphrase responsibly: Use your own words but still cite the source.
- Seek help: If unsure, ask your teachers for guidance.
- Use Plagiarism detection tools.
- Develop your own ideas and perspectives on a topic.

Types of Intellectual Property

a. Patents : A patent is a legal right granted by the government to an inventor. India has a law called the Patents Act of 1970. It protects inventors and encourages new ideas. India gives patent rights in the different areas like new drug formulas, apps and tech tools, new seeds, fertilisers, machines, solar panels or wind turbines.

b. Copyrights

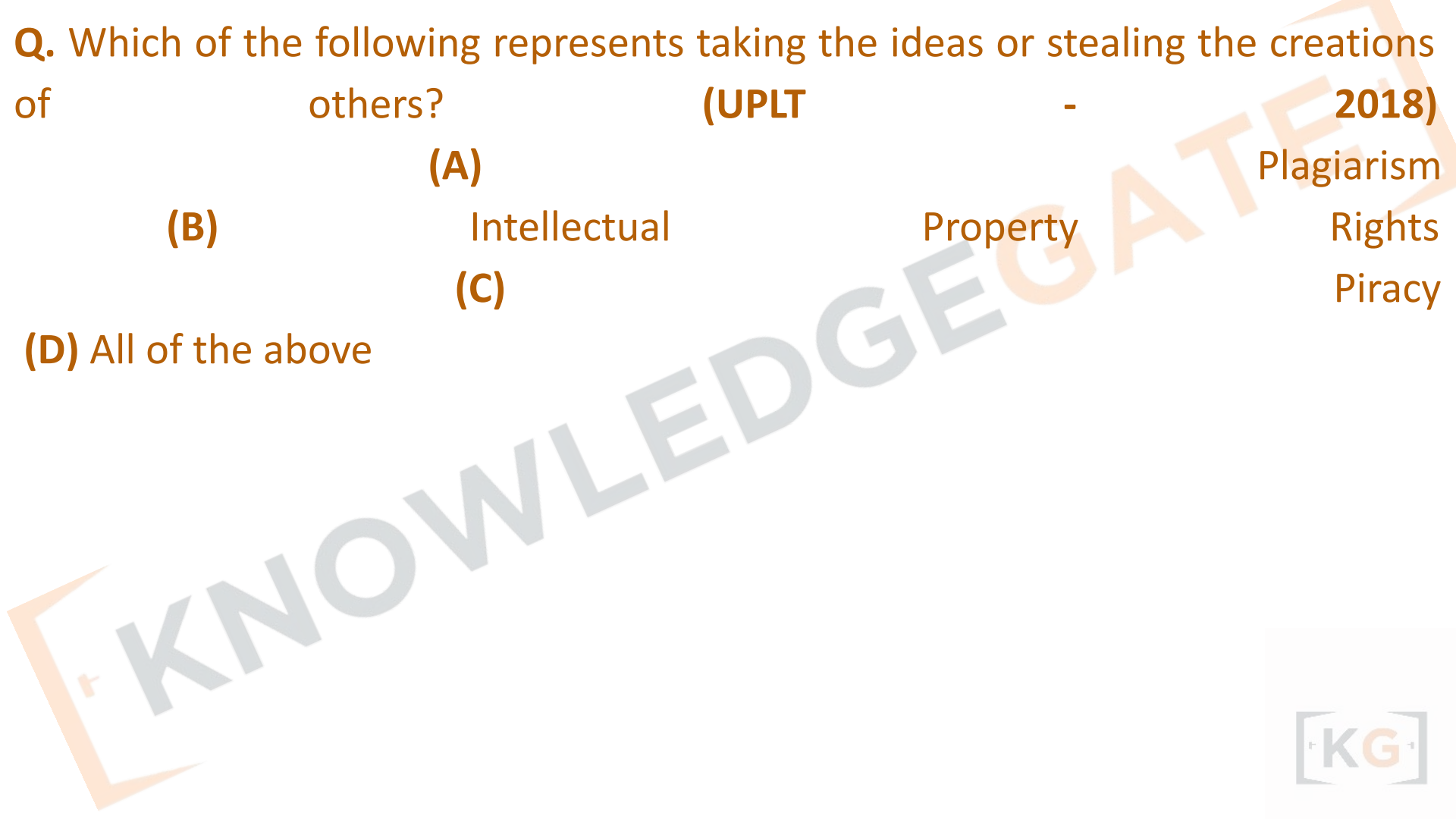
Copyright is a type of intellectual property that protects original creative works. It gives rights to creators of things like books, poems, songs, music, movies, videos, paintings, etc.

This can include various forms such as:

- Literary works: Books, poems, articles, scripts
- Artistic works: Paintings, sculptures, photographs, graphic designs
- Musical works: Songs, compositions
- Dramatic works: Plays, screenplays
- Cinematographic works: Films, videos
- Sound recordings: Recordings of music or spoken words
- Computer programs: Software code

The copyright owner has the exclusive right to:

- Reproduce the work (e.g., copying, printing)
- Prepare derivative works (e.g., creating translations, adaptations)
- Distribute copies of the work (e.g., selling, lending)
- Display the work publicly (e.g., online, in exhibitions)
- Perform the work publicly (e.g., playing music, performing a play)



Q. Which of the following represents taking the ideas or stealing the creations of others? (UPLT - 2018)

- (A) Plagiarism
- (B) Intellectual Property Rights
- (C) Piracy
- (D) All of the above



c. Trademark

A trademark is a type of intellectual property that consists of a recognizable sign, design, or expression that identifies a specific product or service from a particular source and distinguishes it from others.

- Trademarks help consumers recognize the origin of a product or service.
- By establishing clear ownership and rights, trademarks discourage infringement and counterfeiting.
- This allows them to make informed choices based on their past experiences with the brand.
- Trademarks help prevent confusion and ensure consumers are getting what they expect.

Trademarks can take many forms, including:

- Words: Names, slogans, logos, taglines
- Symbols: Logos, icons, designs
- Sounds: Jingles, musical themes

Classification of Cyber-crimes

Crime Against an Individual – Crime committed against an individual by use of an individual's credit card details, confidential data and sending of spam emails. This crime is mainly done for making money.

Crime Against an Organization – This is done against a firm, company or organization to get unauthorized access to the data.

Crime Against Government – This is the crime against the nation, by getting access to the national data and records.

Different Types of Cyber-crimes

- **Cyberbullying:** Cyberbullying is the use of electronic means to bully or harass another person.
- **Cyberstalking:** Cyberstalking is a type of cybercrime that uses the internet and technology to harass or stalk an individual, group, or organization that causes them fear.
- **Phishing:** Phishing is a type of deceptive cybercrime that involves tricking people into providing sensitive information, such as passwords or credit card numbers, by posing as a trustworthy source.
- **Identify Theft:** Identity Theft is a crime where someone steals your personal information, such as your name, Aadhar number, email, phone number, credit card number, or other identifying details, and uses it to commit fraud in your name.

- **Malware Attacks:** Malware Attacks involve the use of malicious software (malware) to harm or exploit a computer system, server, or network. Malware can be delivered through various methods, such as:
 - **Phishing emails:** These emails appear legitimate but contain malicious attachments or links that, when clicked, install malware.
 - **Malicious websites:** Downloading files or visiting compromised websites can unknowingly infect your device with malware.
 - **Infected attachments:** Opening attachments from untrusted sources can introduce malware.
 - **Physical media:** Inserting infected USB drives or CDs can also spread malware.

- **Email spoofing:** Email spoofing is a deceptive tactic where attackers forge the sender address in an email message to make it appear as if it came from someone else.
 - Click on malicious links
 - Open infected attachments
 - Reply to the email and reveal personal information
 - Take unintended actions
- **Denial-of-Service (DoS) Attacks:** These attacks aim to disrupt the normal functioning of a website or server by overwhelming it with traffic, making it unavailable to legitimate users.

- **Ransomware:** Ransomware is a type of malware that prevents or limits users from accessing their system, until a ransom is paid.
- **Hacking:** Hacking refers to the process of gaining unauthorized access to a Computer system or network. This can be with malicious or ethical intentions.
- **Data Breaches:** A data breach occurs when unauthorized individuals gain access to sensitive or confidential information (e.g., financial records, customer's personal information), medical records, passwords etc.

Cyber Crime

Cybercrime refers to criminal activities carried out using computers and the internet, including hacking, data theft, malware attacks, and financial fraud. With businesses, governments, and individuals relying heavily on digital platforms, cyber threats have escalated, leading to billions in financial losses worldwide. In **2023 alone, cybercrime caused over \$8 trillion in damages**, impacting online security on a massive scale. From phishing scams to ransomware attacks, cybercriminals exploit vulnerabilities to steal sensitive information and disrupt systems.

This article explores the **types of cybercrime with examples**, real-world cases, **cybercrime laws**, and **effective prevention measures** to stay protected in the digital world.

What is Cyber Crime?

Cybercrime refers to **illegal activities involving computers, networks, or the internet** as a tool to commit offenses. These crimes include:

- **Identity Theft** – Stealing personal information to commit fraud.
- **Financial Fraud** – Online scams, fake transactions, and credit card fraud.
- **Cyberbullying** – Harassment or threats through digital platforms.
- **Phishing Attacks** – Deceptive emails or websites tricking users into revealing sensitive data.
- **Hacking** – Unauthorized access to systems and data breaches.
- **Malware Attacks** – Spreading viruses, ransomware, and trojans to damage or steal data.

Types of Cyber Crime

Cybercrime includes a **wide range of illegal activities** that exploit computers, networks, and the internet. These crimes can be categorized into **two main types**:

1. Cyber Crimes Targeting Computer Networks or Devices

These crimes involve **direct attacks on computers, servers, or digital infrastructure** to steal data, cause disruption, or damage systems. It involves different threats like **viruses, bugs, etc.** and **(DoS) denial-of-service attacks**.

- **Malware Attacks:** This kind of cyber threat relates to malware viruses, worms, Trojans, etc. for interfering, damaging, or unauthorized access to computer systems.

For example, ransomware encrypts files and then later demands ransom for decryption.

- **Denial-of-Service (DoS) Attacks:** Here, the attackers focus on a system and flood it with high traffic, hence making it inaccessible to the users. Another dangerous variant of DoS is DDoS, wherein many compromised systems target one, thus, much difficult to defend against.

For example, A DDoS attack crashes an e-commerce website by overwhelming its server with traffic.

- **Phishing Attacks:** These are masqueraded e-mails or messages claiming to be from a formal web but only request that the user grant access to sensitive information like password points for an account or credit card numbers. Phishing can be described as an outstanding one of the most common cyber threats.

For example, A fake PayPal login page that steals your credentials.

- **Botnets (Zombie Networks):** A number of hijacked computers can become a "botnet" of malware that can be used by an attacker for coordinated attacks or spamming.

*For example, Hackers use botnets to send **millions of spam emails** in a single day.*

- **Exploits and Vulnerabilities:** The typical area through which cyber-thieves exploit software weakness is the application or operating system vulnerability in order to access it illegally.

For example, Exploiting an outdated banking app to steal user financial details.

2. Crimes Using Computer Networks to Commit Other Criminal Activities

These types of crimes include cyberstalking, financial fraud, or identity theft.

- **Cyberstalking:** This is considered as that crime in the nature of threatening or frightening a person on-line and spreading fear and emotional distress. This can be termed as involving threats, constant monitoring, or receiving repeated unwanted messages.

*For example, Sending **threatening messages** to a person via email or social media.*

- **Financial Fraud:** This is an example of a cybercrook manipulating the victim online to proceed with stealing money, such as fake investment opportunities, hacking a business email, and using someone else's credit card details.

For example, A fake online store that steals credit card details without delivering products.

- **Identity Theft:** It is normally the identity of people whose information is stolen with the intention of only acting like them either to misuse their cash or money from their account or even to do malicious reasons. It always lowers the credit score of the victim and in the worst case scenario, misused the account/loan financially with incorrect transactions.

*For example, A hacker using stolen credentials to **apply for credit cards and loans.***

- **Online Harassment and Hate Crimes:** When people use the internet to discriminate against a particular person based on his or her racial background, gender, religion, or whatever, which can psychologically disturb the harassed person.

For example, Cyberbullying campaigns that target individuals based on race, gender, or religion. **Intellectual Property Theft:** Intellectual property theft refers to the theft of copyrighted content or business secrets through the internet, thereby financially and competitively hurting individuals and companies.

*For example, A software company illegally using another firm's **source code** to create a competing product.*

Examples of Cyber Crime

Cybercrime includes a wide range of illegal activities that exploit the internet, computer systems, and networks for **financial, political, or personal gain**. Here are some of the **most common cybercrime examples**:

1. Cyber Terrorism: Cyber terrorism involves using the internet to **carry out violent threats, disrupt essential services, or spread fear** among people. Cyber terrorists target **critical infrastructure, government systems, or financial institutions** to cause panic or damage.

Example: *Hacking into power grids or communication networks to create widespread disruption*

Examples of Cyber Crime

2. Cyber Extortion (Ransomware Attacks): Cyber extortion happens when hackers attack websites or computer systems and demand money to stop the attacks. They threaten to keep attacking unless they receive a large payment.

Example: *A ransomware attack on a hospital system, blocking access to patient records until a ransom is paid*

3. Cyber Warfare:

Cyber warfare is when countries use computers and networks as part of their battles. It includes both attacking and defending against cyber threats, like hacking and spying.

Example: *A government hacking another country's defense networks to steal classified information.*

4. Internet Fraud:

This type of fraud occurs when someone tricks others on internet to steal money or private information. It involves hiding or giving false information to deceive people and covers many different illegal actions.

Example: *A scam website pretending to sell products but stealing users' payment details instead*

5. Cyber Stalking and Online Harassment

Cyber stalking is a form of online harassment where someone sends threatening messages or emails to a victim they know. If the stalker feels it's not working, they may also start following the victim in real life to make their life more difficult.

***Example:** An ex-partner repeatedly sending threatening messages and tracking a victim's online activity*

6. Financial Fraud:

Cybercriminals steal **personal and financial data** to commit fraud, open fake bank accounts, or make unauthorized transactions. **Phishing attacks** are one of the most common methods used to trick victims into providing sensitive information.

***Example:** A phishing email pretending to be from a bank, asking users to enter their login details on a fake website.*

7. Cyber Espionage:

Cyber espionage refers to **hacking into government agencies, businesses, or corporations to steal confidential data or trade secrets**. It is often used by **competitor businesses or state-sponsored hackers**.

***Example:** A company stealing another firm's product designs through hacking.*

Which of the following would best fit the following description - Software is copied and given to a friend without the permission of the copyright owner?

- (a) freeware
- (b) piracy
- (c) shareware
- (d) public domain

A type of cybercrime that occurs in pay per click online advertising when a person, automated script or computer program imitates a legitimate user of a web browser clicking on an ad, for the purpose of generating charge per click without having actual interest in the targets of the ad's link

- (a) Phishing
- (b) Zombie
- (c) Click fraud
- (d) None of these

RPSC VPITI 2014 (IT)

Ans. (c)



Match the terms with the definition.

(a) Masquerading

(i) session is intercepted

(b) Phishing

(ii) one pretends to be someone else

(c) Hijacking

(iii) a email misleads a user into entering confidential information

Codes:

(a) (b) (c)

(A) (i) (ii) (iii)

(B) (i) (iii) (ii)

(C) (iii) (ii) (i)

(D) (ii) (iii) (i)

Which of the following is a cybercrime

- (a) Hacking
- (b) Worm attack
- (c) Virus attack
- (d) All of them

RPSC VPITI 2014 (IT)

Ans. (d)



Identity theft is the

- (a) impersonation by a thief of someone with a large bank account
- (b) impersonation by a thief of someone with computer skills
- (c) impersonation by a thief of someone with good credit
- (d) Impersonation by as thief of someone's identity for purpose of economic/social gain

An attempt to slow down or stop a computer system or network by flooding the system with requests for information is called a

- (a) virus
- (b) worm
- (c) denial-of-service attack
- (d) Trojan horse

Punjab State Civil Supplies Corp. Ltd. 13.11.2011

Ans. (c)



Challenges of Cyber Crime

- **People are unaware of their cyber rights:** The Cybercrime usually happen with illiterate people around the world who are unaware about their cyber rights implemented by the government of that particular country.
- **Anonymity:** Those who Commit cyber crime are **anonymous** for us so we cannot do anything to that person.
- **Less numbers of case registered:** Every country in the world faces the challenge of cyber crime and the rate of cyber crime is increasing day by day because the people who even don't register a case of cyber crime and this is major challenge for us as well as for authorities as well.
-

Challenges of Cyber Crime

- **Mostly committed by well educated people:** Committing a cyber crime is not a cup of tea for every individual. The person who commits cyber crime is a very technical person so he knows how to commit the crime and not get caught by the authorities.
- **No harsh punishment:** In Cyber crime there is no harsh punishment in every cases. But there is harsh punishment in some cases like when somebody commits cyber terrorism in that case there is harsh punishment for that individual. But in other cases there is no harsh punishment so this factor also gives encouragement to that person who commits cyber crime.

Impact of Cyber Crimes

- **Financial Losses:** The fraud and theft can cause great losses not only for the given organizations but for individuals also.
- **Reputational Damage:** Some people may realize that reputation becomes an issue they may lose depending on the legal outcomes resulting from lawsuits.
- **Operational Disruption:** As will be highlighted later, such an occurrence leads to a shutdown and consequently a loss of productivity.
- **Legal Consequences:** In the cases where clients have been involved in some legal cases or even regulatory fines, they may have to go through another phase of legal activities, clients have to spend considerable amount of money on protecting their data.

How to Protect Yourself Against Cybercrime?

- **Use strong password:** Maintain different password and username combinations for each account and resist the temptation to write them down. Weak passwords can be easily cracked using certain attacking methods like Brute force attack, Rainbow table attack etc, So make them complex. That means combination of letters, numbers and special characters.
- **Use trusted antivirus in devices:** Always use trustworthy and highly advanced antivirus software in mobile and personal computers. This leads to the prevention of different virus attack on devices.
- **Enable Two-Factor Authentication:** Activate two-factor authentication on your accounts for an extra layer of security, which requires a second verification step.
- **Keep your device software updated:** Whenever you get the updates of the system software update it at the same time because sometimes the previous version can be easily attacked.
- **Use secure network:** Public Wi-Fi are vulnerable. Avoid conducting financial or corporate transactions on these networks.

- **Never open attachments in spam emails:** A computer get infected by malware attacks and other forms of cybercrime is via email attachments in spam emails. Never open an attachment from a sender you do not know.
- **Software should be updated:** operating system should be updated regularly when it comes to internet security. This can become a potential threat when cybercriminals exploit flaws in the system.

How to Report a Cybercrime

Reporting cybercrime quickly is important to stop further damage and catch the criminals. Each country has its own way of handling cybercrime, but in most cases, you need to contact local police or a government agency that deals with online crimes.

India: National Cybercrime Reporting Portal: Individuals can report cybercrimes through the official portal at cybercrime.gov.in. This service facilitates the online filing of complaints related to cyber offenses.

Conclusion

Computer criminality forms part of the threats that bring a clear threat increase to personal computer use, businesses, and governments. Enumerating the types of cybercrime, the challenges that are there, and taking preventive measures, one is in a position to minimize the impacts of these threats.

Despite all of this, there are a lot of issues, which remain to be discussed, as the technology advances even more. This simply means that one has to be alert and be sensitive to the new world of threats that are accorded to the cyber world.

Cyber Laws in India

In India, The Information Technology Act, 2000, ALSO aims to provide for the legal framework so that legal sanctity is accorded to all electronic records and other activities carried out by electronic means.

Key Provisions of the Information Technology Act, 2000

- **Legal recognition of electronic records and digital signatures:** The IT Act grants legal validity to electronic documents and signatures, putting them on par with traditional paper-based documents.
- **Data protection and privacy:** The Act includes provisions for safeguarding the privacy of individuals and ensuring the security of sensitive personal data. However, critics argue that these provisions need strengthening.

- **Cybercrime offences:** The IT Act defines various cyber offences, including:
 - Hacking
 - Data theft
 - Identity theft
 - Spreading online obscenity
 - Cyberterrorism
- **Penalties and offences:** The Act outlines punishments for cybercrime offences, including imprisonment and fines.
- **Establishment of the Cyber Appellate Tribunal:** This tribunal was set up to adjudicate disputes arising from the IT Act.

The Information Technology (Intermediary Guidelines and Digital Media Ethics Code) Rules, 2021

These rules provide guidelines for digital intermediaries, social media platforms, and OTT (Over-the-Top) platforms, including requirements related to content removal, grievance redressal mechanisms, and compliance with Indian laws.

The Bharatiya Nyaya Sanhita, 2023 (BNS)

The Bharatiya Nyaya Sanhita, 2023 (BNS) was enacted on December 25, 2023, repealing and replacing the Indian Penal Code, 1860 (IPC) as the new penal code of the country. The three new Laws Bharatiya Nyaya Sanhita, Bharatiya Nagarik Suraksha Sanhita and the Bharatiya Sakshya Adhiniyam came into effect from July 1, 2024. Any crime committed until 30 June 2024 will continue to be governed by the IPC.

Q. The main purpose of the Data Protection Act is to (UPLT - 2018)

- (A) Protect personal privacy**
- (B) Prevent viruses**
- (C) Increase the security of computer systems**
- (D) Reduce project failures**

